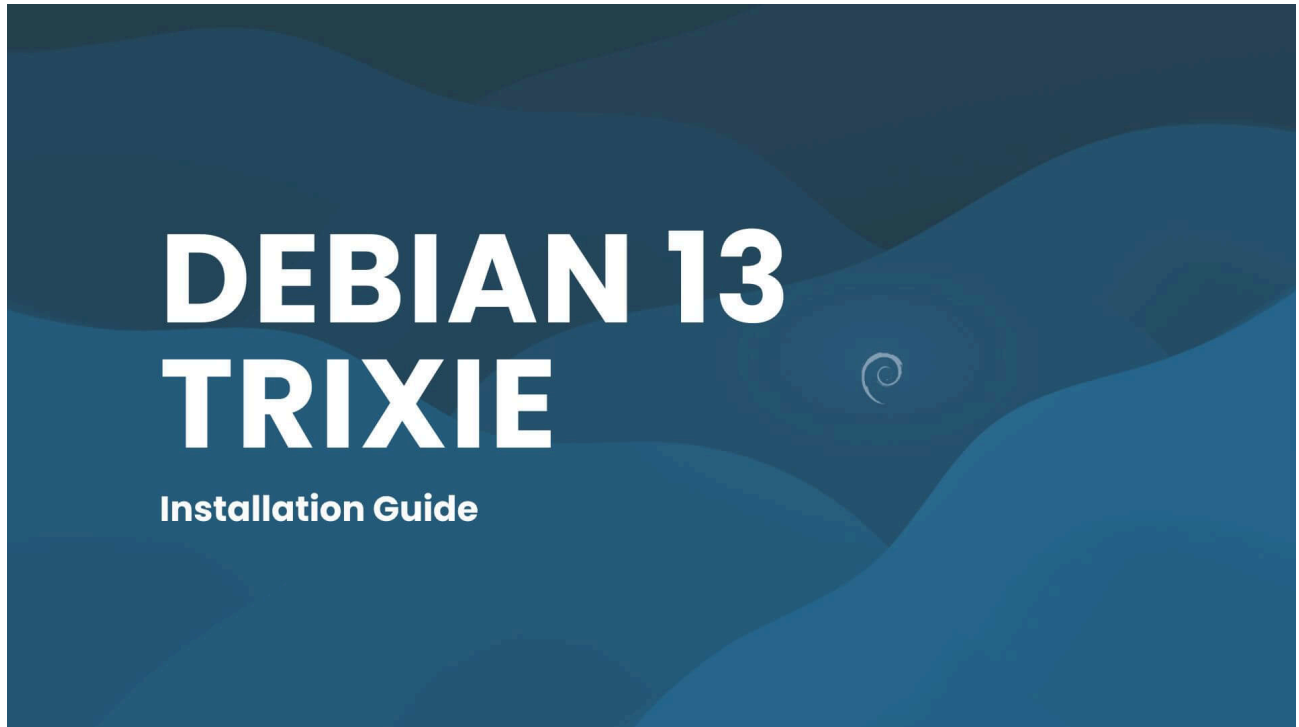


How To Install Debian 13 Trixie Desktop [A Step-by-Step Guide]

 ostechnix.com/install-debian-13-trixie-desktop

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August 10, 2025



A Detailed Step-by-Step Guide to Setup Debian 13, Codenamed Trixie.

Published: Updated: 8433 views 18 mins read

Welcome to this comprehensive guide on **installing Debian 13**, the latest release codenamed **Trixie**! Debian 13 brings a multitude of enhancements, including updated software packages, improved hardware support, and increased stability. This step-by-step tutorial will walk you through the steps to **install Debian 13 Trixie**, ensuring a smooth and successful setup of Debian 13 on your system.

Debian 13 Release Notes

Wondering what's new in Debian 13 trixie edition? Refer our **Debian 13 Release Notes** guide in the link below.

[Debian 13 Bookworm Released!](#)

If you're already running Debian 12 on your system, follow our Step-by-Step Guide to upgrade to Debian 13 Trixie from Debian 12 Bookworm.

[How To Upgrade To Debian 13 Trixie From Debian 12 Bookworm](#)

To Perform a Fresh Installation of Debian 13, Follow the Steps Below.

Download Debian 13

When it comes to Debian 13, you have a choice of three different image types to suit your installation needs:

1. **Netinstall** (small installation image): This image requires an internet connection during installation. It provides a minimal setup and allows you to download packages from the Debian repositories as needed.
2. **Cloud images**: These images are designed for deploying Debian on various cloud platforms such as OpenStack, Amazon EC2, and Microsoft Azure. They are specifically optimized for cloud environments.
3. **Complete installation image**: If you're looking to set up a comprehensive Debian desktop or server on a bare metal system, this is the image you'll want. It contains all the necessary packages and components for a full-fledged Debian experience.

To get started, head over to the [official Debian download page](#) and locate the Debian 13 ISO image that matches your machine's architecture. Once you've chosen the appropriate image, proceed to download it.

Debian provides various other installers and images to accommodate different needs. If you require a live system, an offline installer for systems without a network connection, an installer for alternative CPU architectures, or even a Debian image for cloud instances, you can find them at [Getting Debian link](#).

The following table includes the direct Debian 13 download links for various formats:

Please note that these links are subject to change and it's always a good idea to go to the [official Debian website](#) to get the latest download links.

Starting with the release of Debian 12 bookworm, firmware is now included in the regular Debian installer images. This means that users will no longer have to search for special versions or sources to obtain the necessary firmware.

Debian has made the process even more convenient by integrating the firmware directly into the standard installer, eliminating the need for additional steps or downloads.

If you plan to install Debian 13 on a bare metal system, you'll need to create a bootable USB drive. In the following section, you'll find a variety of methods and tools listed that can help you create a bootable Debian Linux USB drive.

Create Debian 13 Bootable USB Medium

Once you have successfully downloaded the Debian 13 ISO image, the next step is to create a bootable USB medium.

Fortunately, there are several reliable tools available for this purpose. You can choose from a range of options listed below, all of which can assist you in creating a bootable USB for Debian 13, codenamed Trixie.

Command line bootable USB creation tools:

- [How To Create Multiboot USB Drives With Ventoy In Linux](#)
- [How To Create Bootable USB Drive Using dd Command](#)
- [Bootiso Lets You Safely Create Bootable USB Drive In Linux](#)

Graphical bootable USB creation tools:

- [Create Bootable USB Drive With Ventoy WebUI In Linux](#)
- [Create Bootable USB Drives And SD Cards With Etcher In Linux](#)
- [Popsicle – Create Multiple Bootable USB Drives At Once](#)
- [Create Bootable USB Drive With USBImager In Linux](#)
- [Kindd – A Graphical Frontend To dd Command](#)

If you intend to install and test Debian on a hypervisor such as VirtualBox, KVM, or Proxmox, there's no need to create a bootable media. Instead, you can directly boot up the Debian 13 ISO image within your virtualization software.

Debian 13 Installation Step-by-Step

Ensure that your system's BIOS is configured to prioritize USB as the primary bootable device. Then, boot your system using the freshly created Debian 13 bootable media and follow the steps below to proceed with the installation of Debian 13, codenamed Trixie.

STEP 1 : Choose Install Type

From the Debian installer menu, choose the option labeled **"Graphical Install"** to initiate the installation process of Debian 13.



Choose Install Type from Debian 13 Installer Wizard

STEP 2: Language Selection

In this step, you will be prompted to select the language that will be used for both the installation process and the system language in Debian 13.

Choose your preferred language from the available options to proceed with the installation.



Choose Installation and System Default Language in Debian 13

STEP 3: Select Your Location

In this step, you will be required to select your location, which will automatically set the appropriate time zone for your Debian 13 installation.

© debian 13

Select your location

The selected location will be used to set your time zone and also for example to help select the system locale. Normally this should be the country where you live.

This is a shortlist of locations based on the language you selected. Choose "other" if your location is not listed.

Country, territory or area:

India

Ireland

Israel

New Zealand

Nigeria

Philippines

Seychelles

Singapore

South Africa

United Kingdom

United States

Zambia

Zimbabwe

other

Screenshot

Go Back

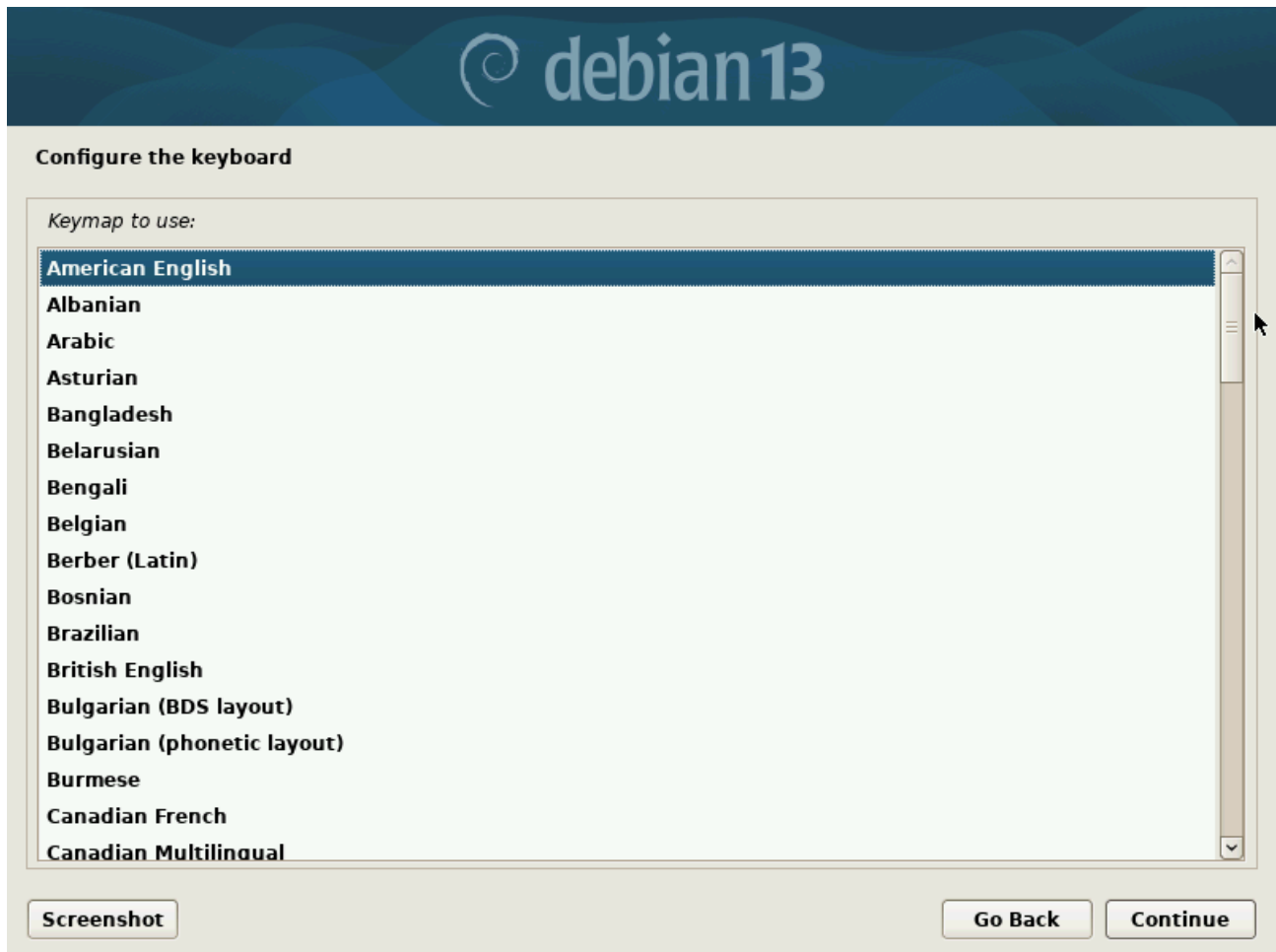
Continue

Select Your Location

STEP 4: Keyboard Configuration

Select the appropriate keyboard layout from the available options that matches your physical keyboard and click Continue.

6/57



Choose the Keyboard Layout

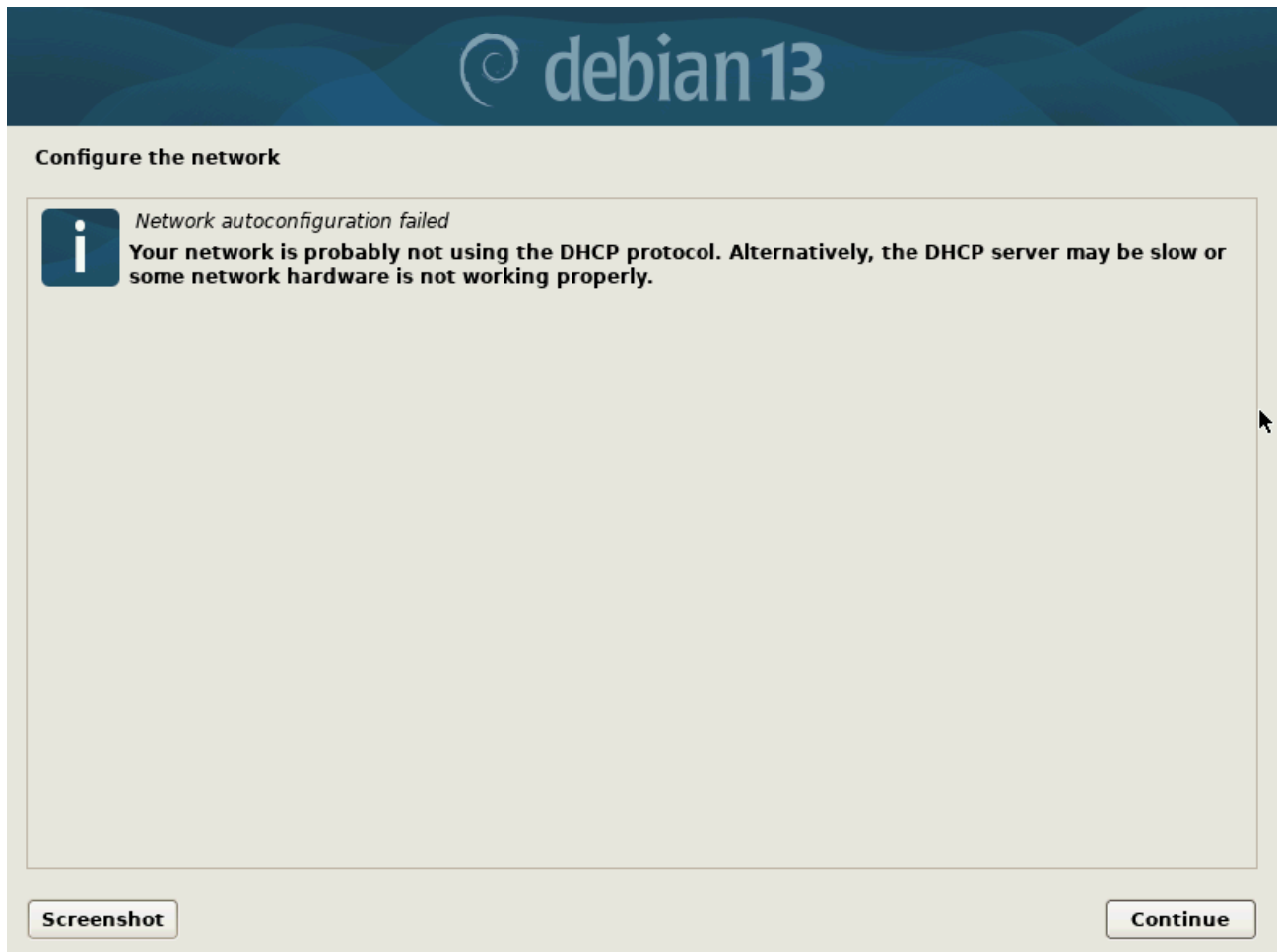
After choosing Keyboard layout, click Continue. Now, some installer components will be loaded from the disk before going to the next step.

STEP 5: Network Configuration

During the installation process, the Debian installer will attempt to automatically configure the IP address by obtaining it from an existing DHCP server in your network.

If no DHCP server is available, the automatic network autoconfiguration will fail. In such a scenario, you will need to manually configure the IP address settings.

Click Continue to manually configure your network.



Configure the Network

In the next screen, choose "Configure network manually" option and click Continue.



Select Configure Network Manually Option

Enter a IP address to your Debian system and click Continue.

Configure the network

The IP address is unique to your computer and may be:

- * four numbers separated by periods (IPv4);
- * blocks of hexadecimal characters separated by colons (IPv6).

You can also optionally append a CIDR netmask (such as "/24").

If you don't know what to use here, consult your network administrator.

IP address:

Screenshot

Go Back

Continue

Enter the IP Address

Configure the network

The netmask is used to determine which machines are local to your network. Consult your network administrator if you do not know the value. The netmask should be entered as four numbers separated by periods.

Netmask:

[Screenshot](#)[Go Back](#)[Continue](#)

Enter Netmask

Enter Gateway IP and click Continue.



Configure the network

The gateway is an IP address (four numbers separated by periods) that indicates the gateway router, also known as the default router. All traffic that goes outside your LAN (for instance, to the Internet) is sent through this router. In rare circumstances, you may have no router; in that case, you can leave this blank. If you don't know the proper answer to this question, consult your network administrator.

Gateway:

192.168.1.101

Screenshot

Go Back

Continue

Enter Gateway Address

Enter Name server (DNS) address and click Continue. Please note that if you want to use multiple DNS addresses, specify them separated by spaces. Do not use commas.

Configure the network

The name servers are used to look up host names on the network. Please enter the IP addresses (not host names) of up to 3 name servers, separated by spaces. Do not use commas. The first name server in the list will be the first to be queried. If you don't want to use any name server, just leave this field blank.

Name server addresses:

1.1.1.1 8.8.8.8

Screenshot

Go Back

Continue

Enter DNS Server Address

Enter a hostname for your Debian system and click Continue.



Configure the network

Please enter the hostname for this system.

The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.

Hostname:

Screenshot Go Back Continue

Enter a Hostname for Debian 13 Desktop

In this step, enter a Domain name for your Debian system. The domain name helps identify and categorize systems within a network.

If you have a specific domain name associated with your network, enter it here. Otherwise, you can leave this field blank.

Providing a domain name is optional, and it can be configured or modified later if needed.



Configure the network

The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.

Domain name:

Screenshot

Go Back Continue

Enter Domain Name

STEP 6: Set Root Password

In this step, set a strong and unique password for the **root** user account.



Set up users and passwords

Some account needs to be available with administrative super-user privileges. The password for that account should be something that cannot be guessed.

To allow direct password-based access via the 'root' account, you can set the password for that account here.

Alternatively, you can lock the root account's password by leaving this setting empty, and instead use the system's initial user account (which will be set up in the next step) to gain administrative privileges. This will be enabled for you by adding that initial user to the 'sudo' group.

Note: what you type here will be hidden (unless you select to show it).

Root password:

☐ Show Password in Clear

Please enter the same root password again to verify that you have typed it correctly.

Re-enter password to verify:

☐ Show Password in Clear

Screenshot

Go Back

Continue

Set Root User Password

A good password should meet the following criteria:

- **Length:** It should have a minimum length of eight characters or more.
- **Complexity:** It should include a combination of uppercase and lowercase letters, numbers, and special characters (@, #, \$, etc.).
- **Uniqueness:** Avoid using common or easily guessable passwords. Do not reuse passwords from other accounts.
- **Avoid personal information:** Do not include personal details such as your name, birthdate, or any other easily identifiable information.

STEP 7: Create New User Account

Using the root user for all activities is not considered a best practice in terms of system security. The root account should be reserved solely for system-related tasks and activities that requires elevated privileges.

So, it is recommended to create a new user account during this step. This new user account will serve as an alternative to the root user, enabling you to perform regular activities without the need for administrative permissions.

Enter a full name for the new user.

Set up users and passwords

A user account will be created for you to use instead of the root account for non-administrative activities.

Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.

Full name for the new user:

[Screenshot](#)[Go Back](#)[Continue](#)

Enter Full Name for the New User

Enter a username for the new user account. The username should start with a lower-case letter, followed by any combination of numbers and more lower-case letters.



Set up users and passwords

Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.

Username for your account:

[Screenshot](#) [Go Back](#) [Continue](#)

Enter Username for the New Account

Please note that hostname, full name and username need not to be the same. You can either use the same name for all or different names for each.

Enter a unique and strong password for the new user. Ensure the user's password is different than the root password.



Set up users and passwords

Make sure to select a strong password that cannot be guessed.
Choose a password for the new user:

●●●●●●

☐ Show Password in Clear

Please enter the same user password again to verify you have typed it correctly.
Re-enter password to verify:

●●●●●●

☐ Show Password in Clear

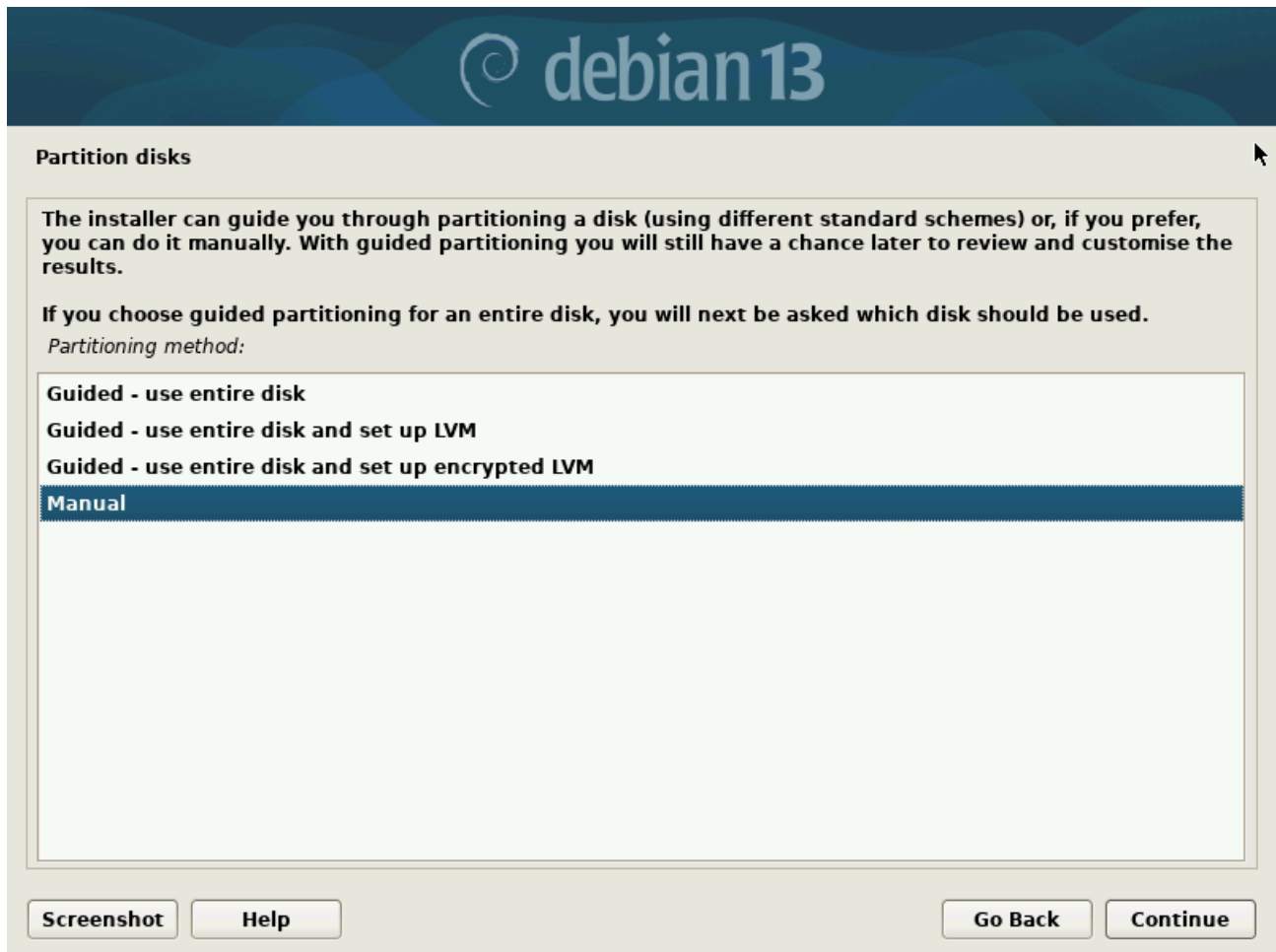
[Screenshot](#) [Go Back](#) [Continue](#)

Set User Password

In this step, you will need to decide how you want to partition your disks in Debian 13. You have four options to choose from:

1. : Selecting this option allows the system to automatically partition the entire disk for you. If you're newbie, choose this option. The installer will setup partitions automatically.
2. : This option automatically sets up Logical Volume Management (LVM) in addition to partitioning the entire disk. LVM provides flexibility in managing storage volumes.
3. **Guided Partition - Use entire disk and set up encrypted LVM:** If you require an extra layer of security, this option will automatically set up an encrypted LVM configuration along with partitioning the entire disk.
4. **Manual partitioning:** If you prefer to have complete control over the disk partitioning process, you can choose manual partitioning. This option allows you to customize and define the partitions according to your specific requirements.

Here, I am choosing **Manual** partitioning method.



Choose Disk Partitioning Method

You will now see an overview of currently configured partitions and mount points. You can choose an existing partition to modify it, or a free space to create new partitions, or disk drive to initialize a new partition table.



Partition Disks

After choosing the device, choose "Yes" to proceed with creating new partition table on the selected device. This action will initialize the disk with a fresh partition table, allowing you to define and allocate partitions as per your requirements.



Create a New Partition Table

Once the empty partition table has been successfully created, you can proceed by selecting the available free space on the disk to create new partitions.

By choosing the free space, you can define and allocate the necessary partitions based on your specific needs and preferences.

This step allows you to efficiently organize and utilize the available disk space by creating partitions that will serve various purposes, such as root (/), home (/home), swap, or any other desired partition layout.



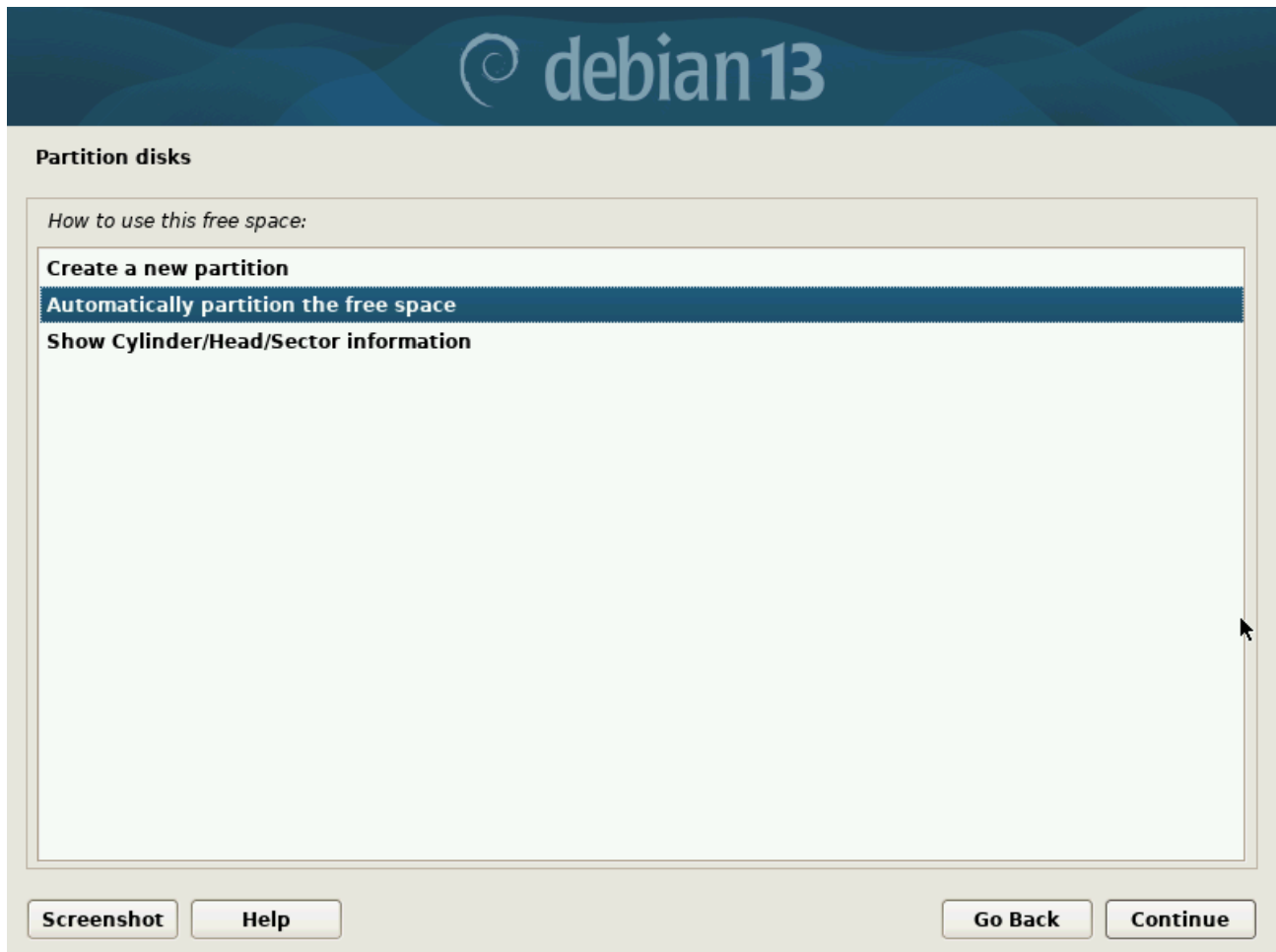
Select Free Space to Create New Partitions

In this step, you have the option to decide how to utilize the available free space on the disk. You will be presented with two choices:

1. **Create a new partition:** By selecting this option, you can manually define and configure individual partitions within the available free space. This allows for greater customization and control over the partitioning process, enabling you to allocate specific sizes and mount points for each partition according to your requirements.
2. **Automatically partition the free space:** If you prefer a more automated approach, choosing this option will prompt the installer to automatically partition the free space on your behalf. The installer will utilize a predefined partitioning scheme or algorithm to allocate the free space based on the recommended configuration.

I have opted for the automated approach by selecting **option 2**. This choice allows Debian 13 installer to automatically partition the available free space on the disk, eliminating the need for manual intervention.

By selecting the option 2, the installer will utilize a predefined partitioning scheme or algorithm to efficiently allocate the free space based on the recommended configuration. This automated approach saves time and simplifies the partitioning process during the installation of Debian 13.



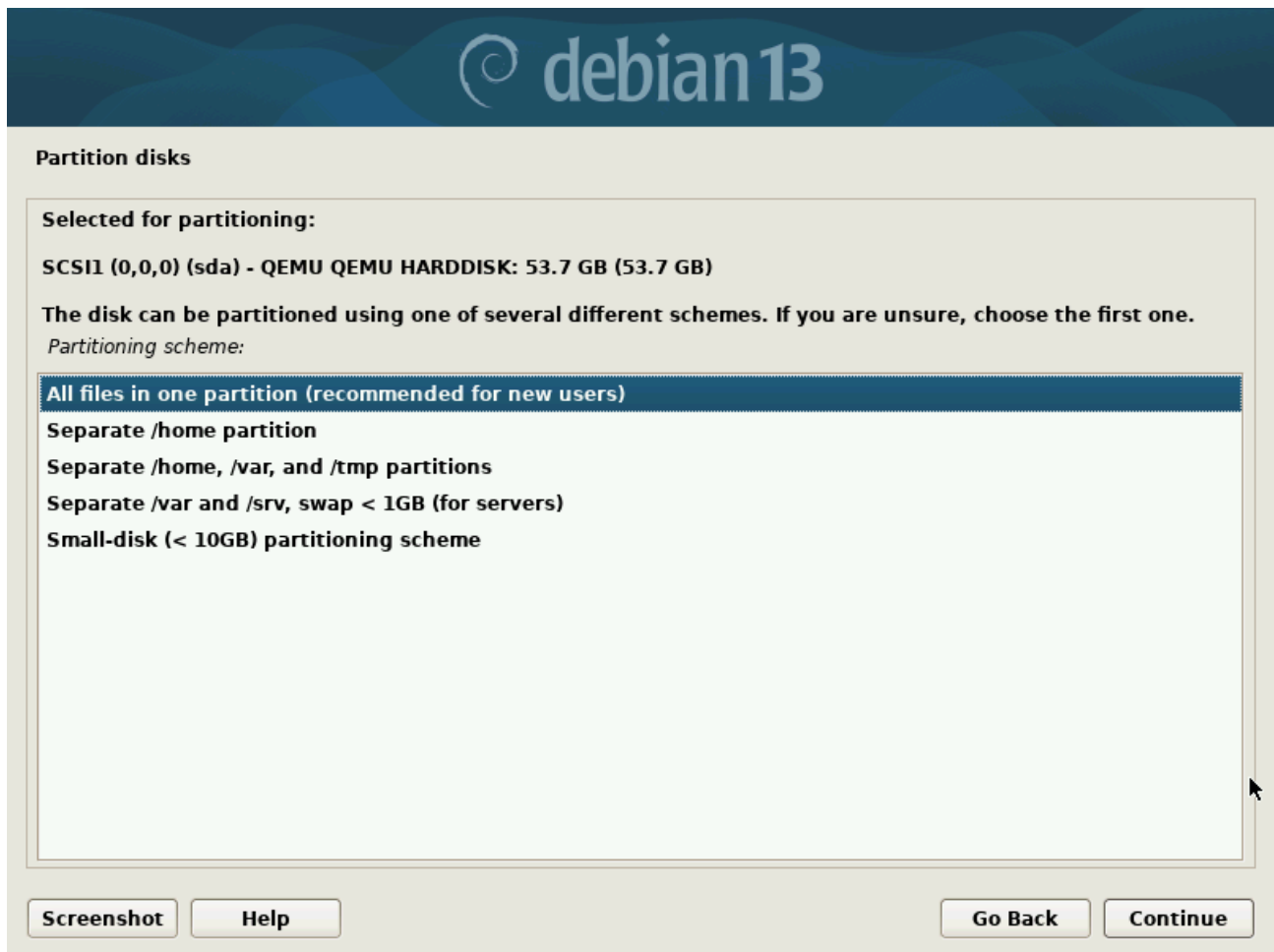
Choose Automatically Partition the Free Space Option

Once you have selected the "Automatically partition the free space" option, the next step is to choose your preferred partitioning scheme. You will be presented with five options to choose from:

1. **All files in one partition** (recommended for new users): This option combines all files, including the root directory, user data, and system files, into a single partition. It is a straightforward and convenient choice, especially for new users who prefer simplicity and ease of use.
2. **Separate /home partition**: By selecting this option, Debian 13 will create a separate partition for the /home directory. This allows for a clear separation of user data from system files, making it easier to manage and maintain personal files.
3. **Separate /home, /var, and /tmp partitions**: This option takes partitioning a step further by creating separate partitions for the /home, /var, and /tmp directories. Each partition serves a specific purpose: /home stores user data, /var contains variable data such as logs and databases, and /tmp is used for temporary files. This approach provides increased flexibility and isolation for different types of data.
4. **Separate /var and /srv, swap < 1GB for servers**: This option creates separate partitions for /var and /srv to isolate log files and service data and limits swap space to under 1 GB.

5. : This option configures Debian to fit within . Ideal for dmbdedded devices, minimal virtual machines, systems with very small storage (e.g., <10 GB disks or USB drives).

Choose the partitioning scheme that best suits your needs and preferences. I go with the **1st option** i.e. **All files in one partition**.



Choose Partitioning Scheme

Based on your previous selection of the partitioning scheme, the Debian installer will automatically create the required partitions for you.

Once the partitions have been set up, choose the option and click on "Continue" to proceed to install Debian 13 Trixie.

Partition disks

This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.

Guided partitioning

Configure software RAID

Configure the Logical Volume Manager

Configure encrypted volumes

Configure iSCSI volumes

▽ SCSI1 (0,0,0) (sda) - 53.7 GB QEMU QEMU HARDDISK

>	#1	primary	50.9 GB	f	ext4	/
>	#5	logical	2.8 GB	f	swap	swap

Undo changes to partitions

Finish partitioning and write changes to disk

Screenshot

Help

Go Back

Continue

Finish Partitioning and Write Changes to Disk

Choose "Yes" to confirm writing partition changes to disk and click Continue.

Partition disks

If you continue, the changes listed below will be written to the disks. Otherwise, you will be able to make further changes manually.

The partition tables of the following devices are changed:
SCSI1 (0,0,0) (sda)

The following partitions are going to be formatted:
partition #1 of SCSI1 (0,0,0) (sda) as ext4
partition #5 of SCSI1 (0,0,0) (sda) as swap

Write the changes to disks?

☐ No

☒ Yes

Screenshot

Continue

Confirm Writing Changes to Disk

The Debian 13 base installation will start now.

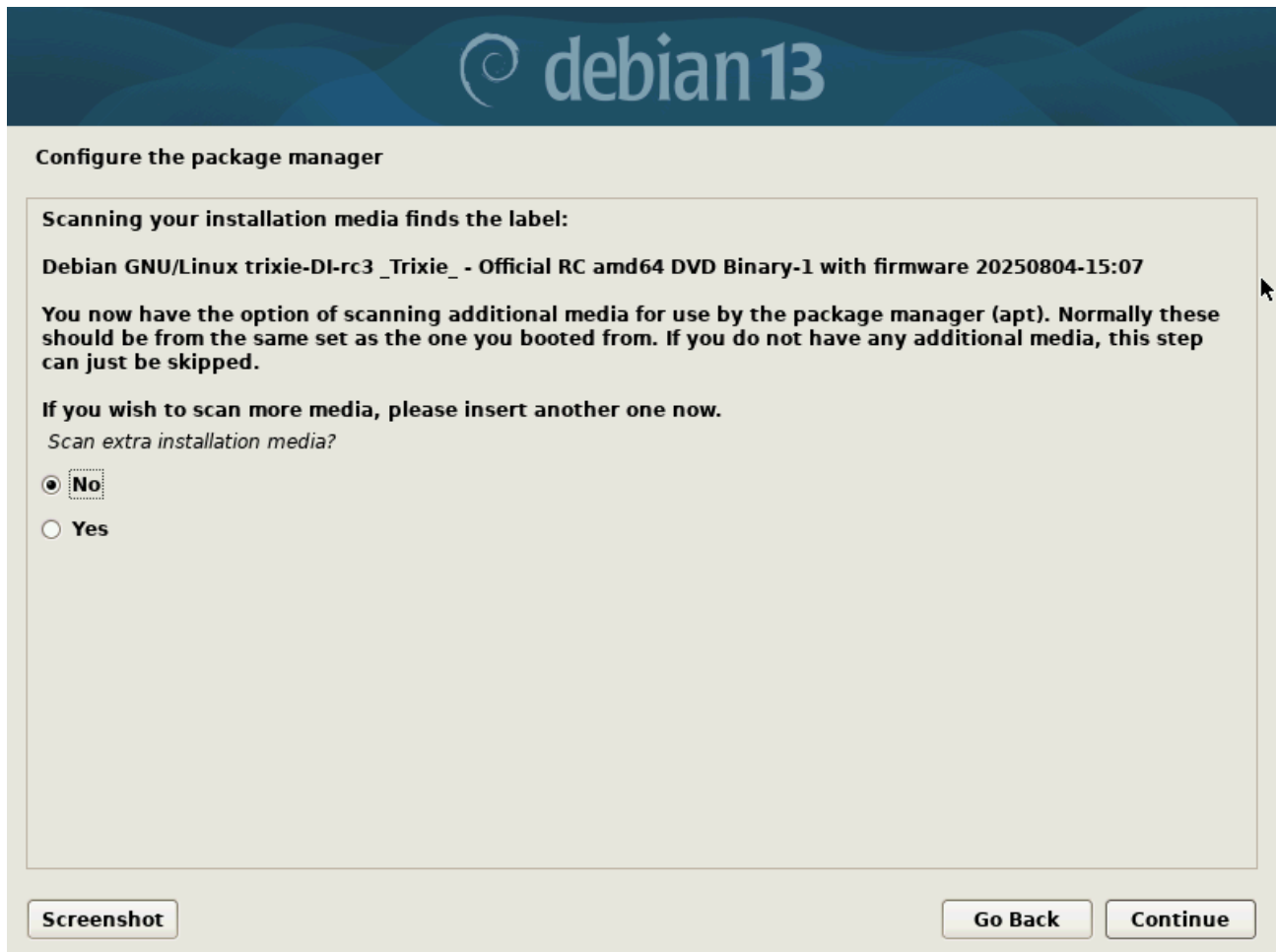


Installing Debian 13 Base System

STEP 9: Configure Package Manager

After a few minutes, the installer will prompt you to scan for additional media. This scan is primarily performed to identify any supplementary media that can be utilized by the package manager (apt).

In most cases, you will not have any additional media apart from the ISO image you initially booted from. Therefore, simply choose the **"No"** option to proceed to the next step.



Scan Extra Installation Media

Next, you will be prompted to configure the network mirror. Configuring network mirrors ensures that the latest packages are installed using the Internet, even though a comprehensive set of packages is included in the installation DVD media (ISO).

If you have an Good internet connection available, it is recommended to choose **"Yes"**. This allows Debian 13 to fetch and install the most up-to-date packages from the network mirror during the installation process, ensuring that you have the latest software versions and security updates.

But, if you do not have an internet connection at the moment or prefer to install using only the packages included in the ISO, select **"No"**. This option will proceed with the installation using the packages available on the installation media, without accessing the network mirror.

Consider your specific circumstances and requirements to make the appropriate choice between "Yes" and "No" during this step of the installation process.

Configure the package manager

A network mirror can be used to supplement the software that is included on the installation media. This may also make newer versions of software available.

You are installing from a DVD image. Even though it contains a large selection of packages, some may be missing. If you have a reasonably good Internet connection, use of a mirror is suggested if you plan to install a graphical desktop environment.

Use a network mirror?

- ☐ No
☒ Yes

Screenshot

Go Back

Continue

Use Network Mirror

If you chose "Yes" in the previous step to configure the network mirror, the installer will now prompt you to select a nearby mirror from a list of countries. Please note that choosing mirrors from nearby countries or even your own, may not always be the best choice.



Choose Nearest Debian Archive Mirror

Select a mirror from the list of available Debian archive mirrors and click Continue.



Select a Debian Archive Mirror

If your machine is configured to access the internet through a proxy, you will be prompted to enter the proxy information in this step.

Provide the necessary proxy details, including the proxy server address and port number. If you do not have a proxy configured or do not require a proxy for internet access, you can leave the fields blank and proceed by pressing the "Continue" button.



Configure the package manager

If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[user][:pass]@host[:port]/".
HTTP proxy information (blank for none):

Screenshot

Go Back

Continue

Enter HTTP Proxy Server Details

STEP 10: Popularity Contest Configuration

In this step, you will be given the option to participate in a popularity contest. By choosing "Yes", you consent to sending package statistics to the developers on a weekly basis. This information assists the developers in understanding package usage patterns and improving software development.

If you wish to participate in the popularity contest, select "Yes". If you prefer to skip participation, choose "No".



Popularity Contest Configuration

You can always modify your choice later by running the following command from your Terminal:

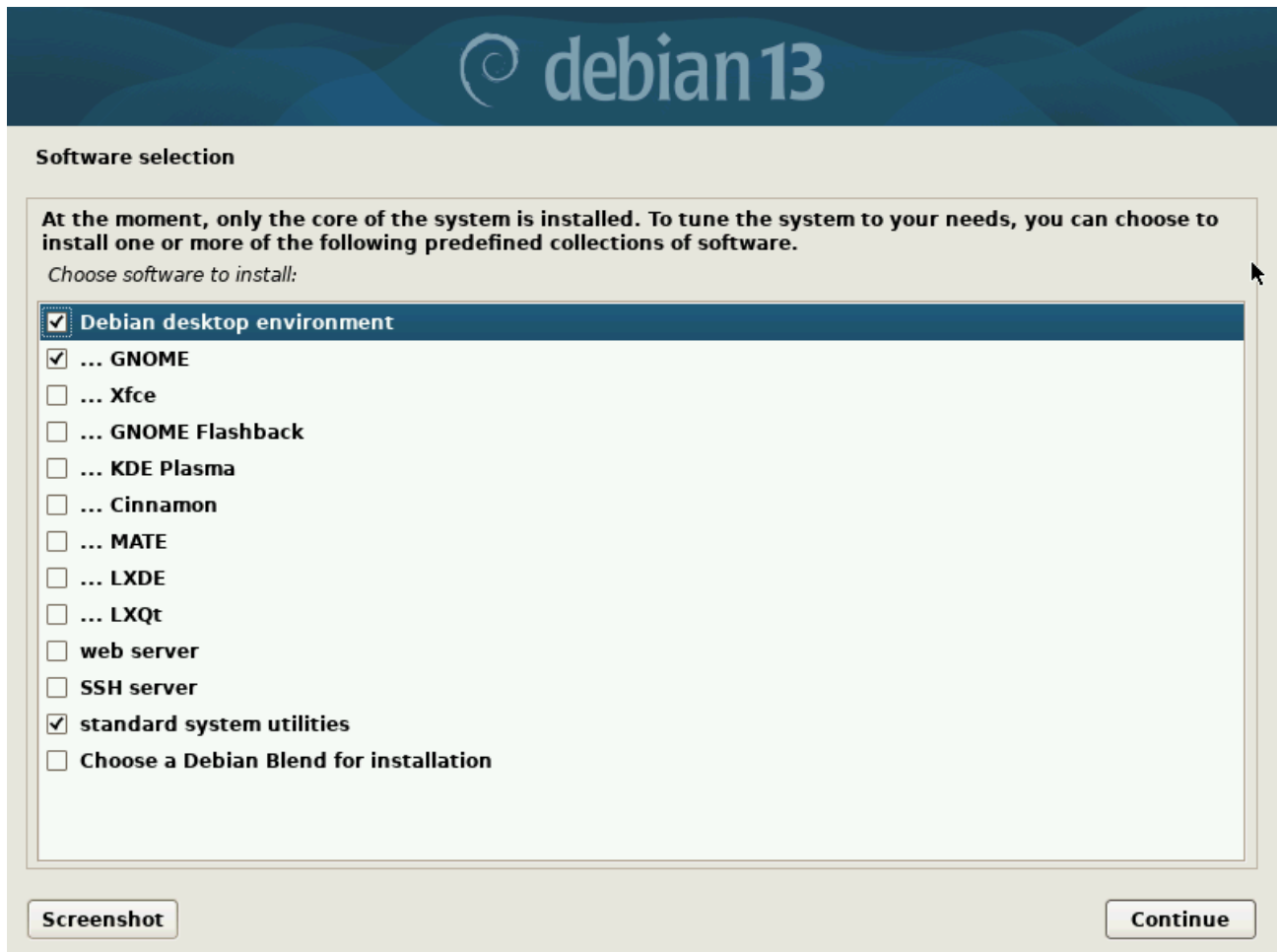
```
dpkg-reconfigure popularity-contest
```

This command allows you to reconfigure your choice regarding the popularity contest at any time after the Debian 13 installation.

STEP 11: Software selection

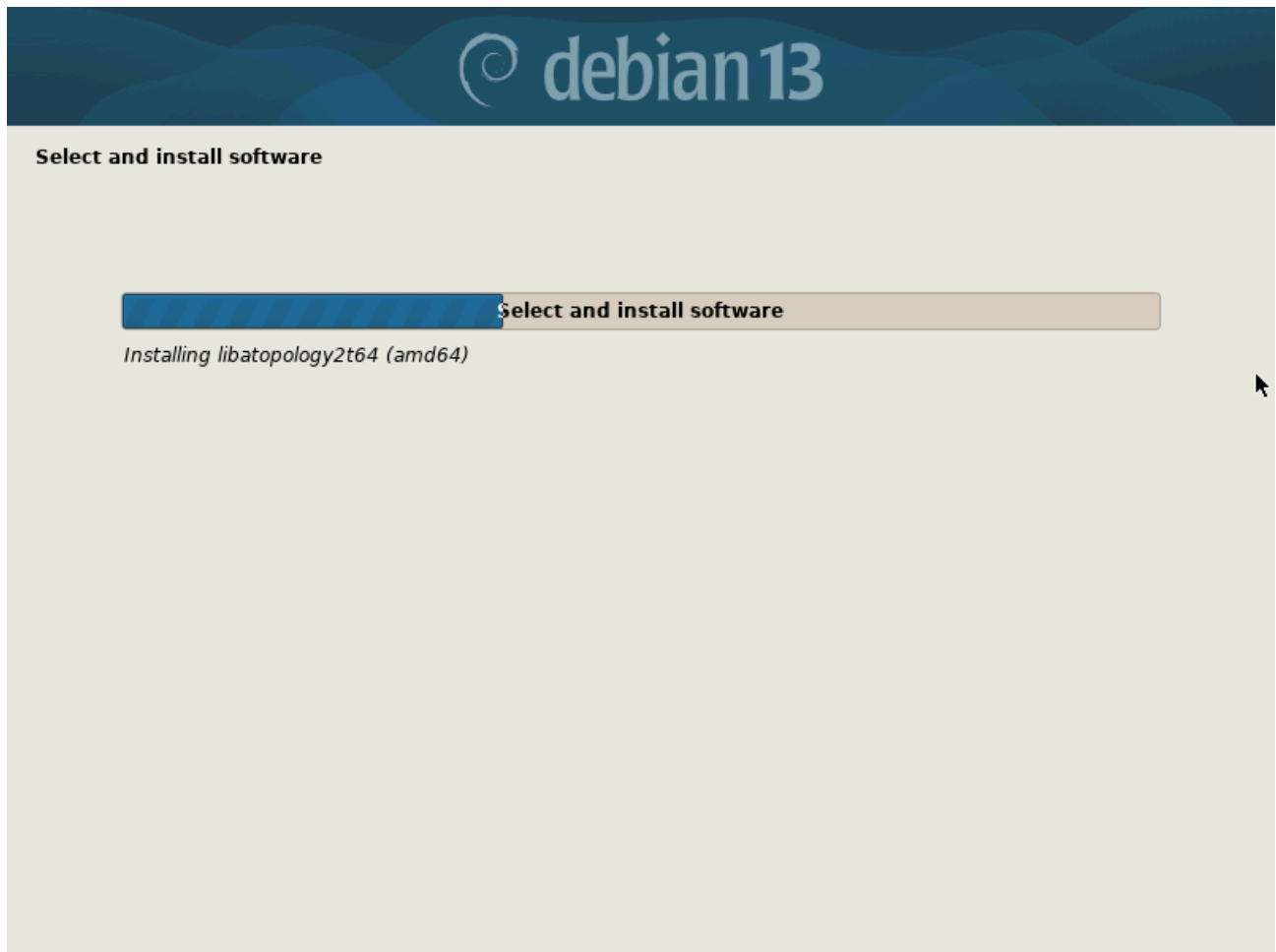
Currently, only the core components of the system have been installed. To customize the system according to your specific requirements, you have the option to install one or more predefined collections of software.

Choose your preferred software to install and click Continue. For example, I am selecting to install GNOME desktop environment and standard system utilities.



Choose Software to Install

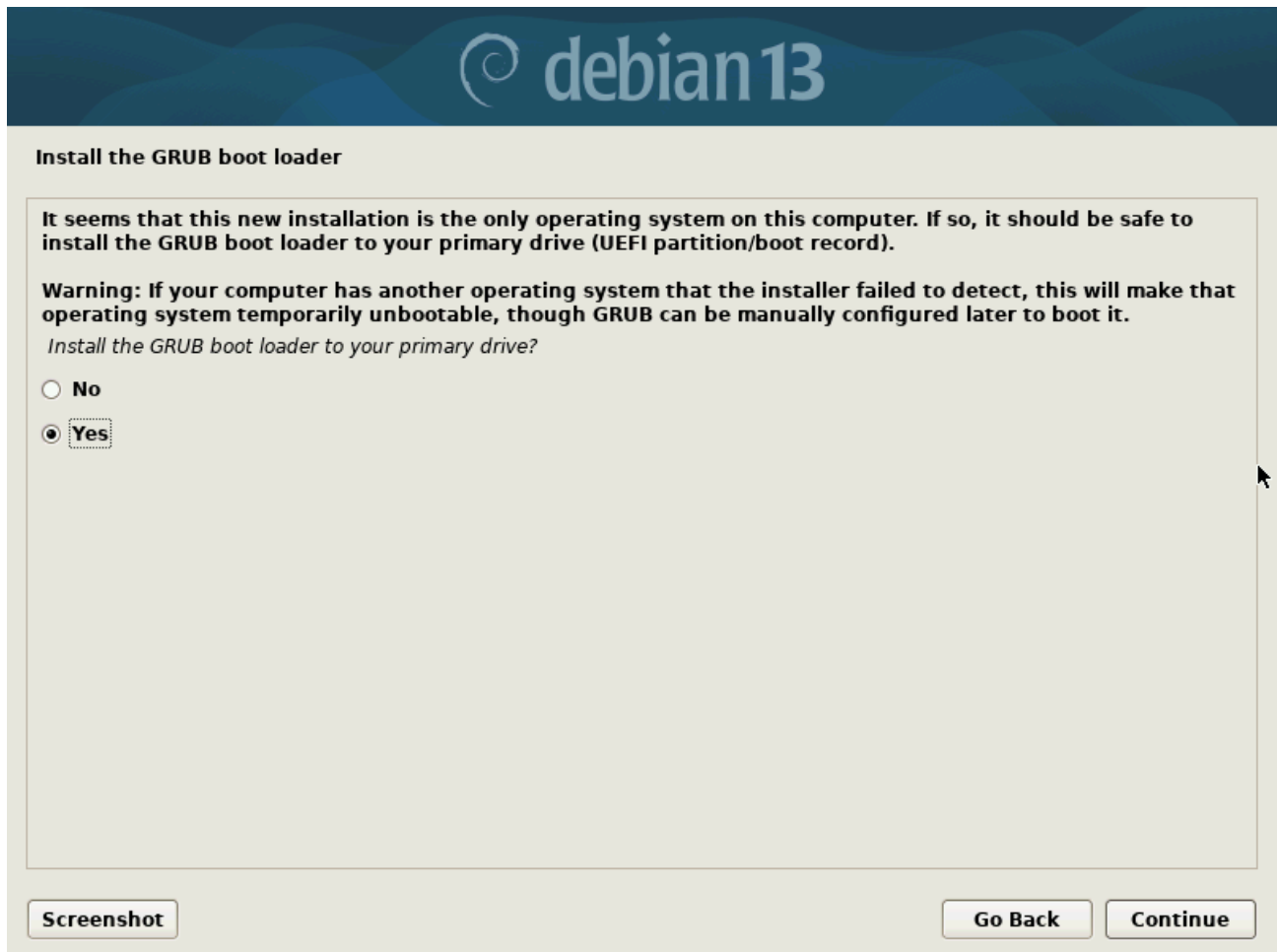
Now the installer will start installing the selected software.



Debian 13 Trixie Software Installation

STEP 12: GRUB Installation

In this step, you will be prompted to install the GRUB boot loader on the primary drive of your system. To proceed with the installation of GRUB, choose "Yes" and press the "Continue" button.



Install the GRUB Boot Loader

Select the drive on which you want to install the boot loader and click Continue.



Select the Device for Boot Loader Installation

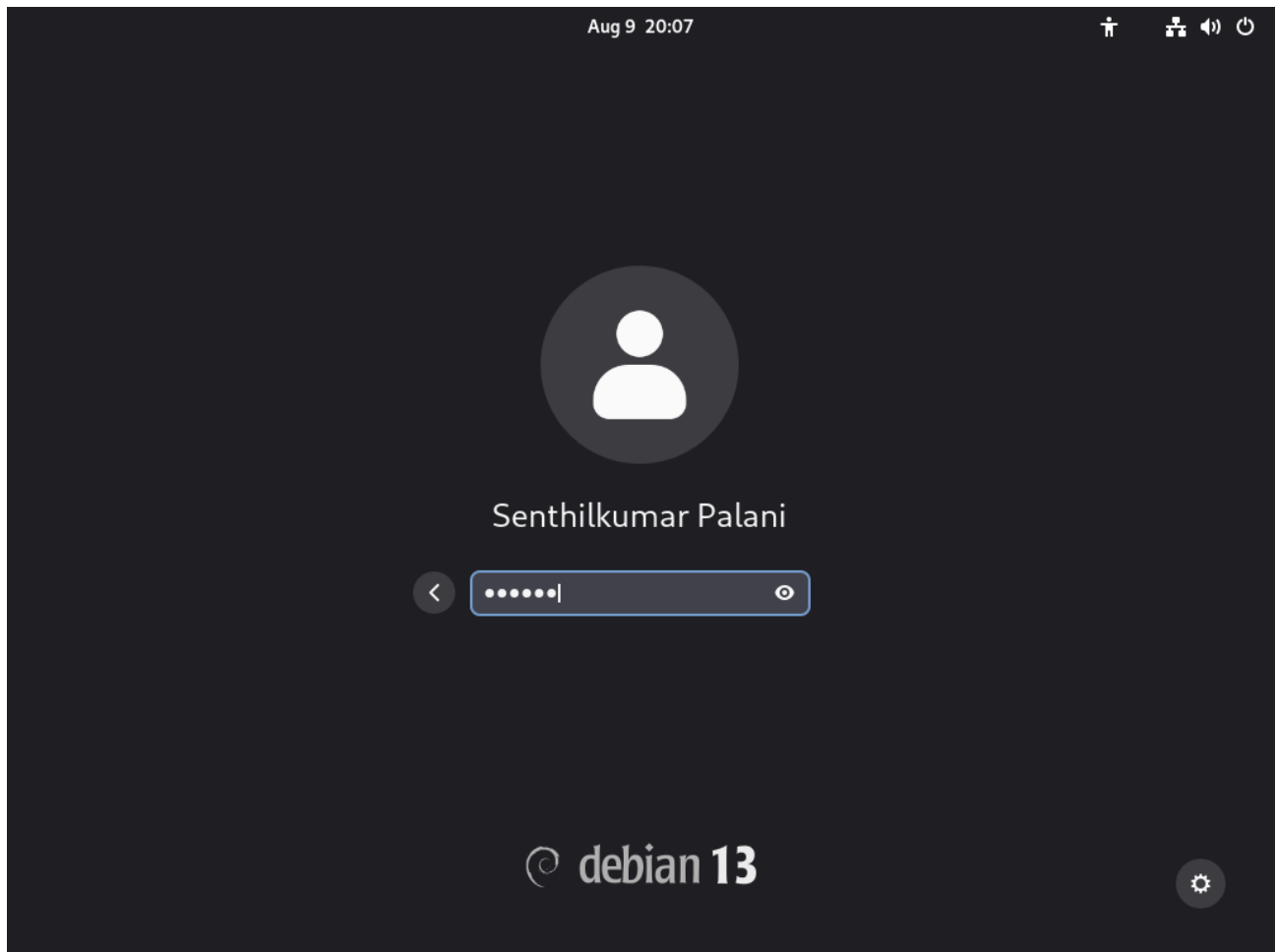
Now, the installer will proceed to install the boot loader. This process may take a few moments.

Once the boot loader installation is complete, proceed to the next step by pressing the "Continue" button.



Debian 13 Trixie Installation is Completed

After the installation process is complete, your system will be rebooted. At the login window, enter the user's password you created during the installation.

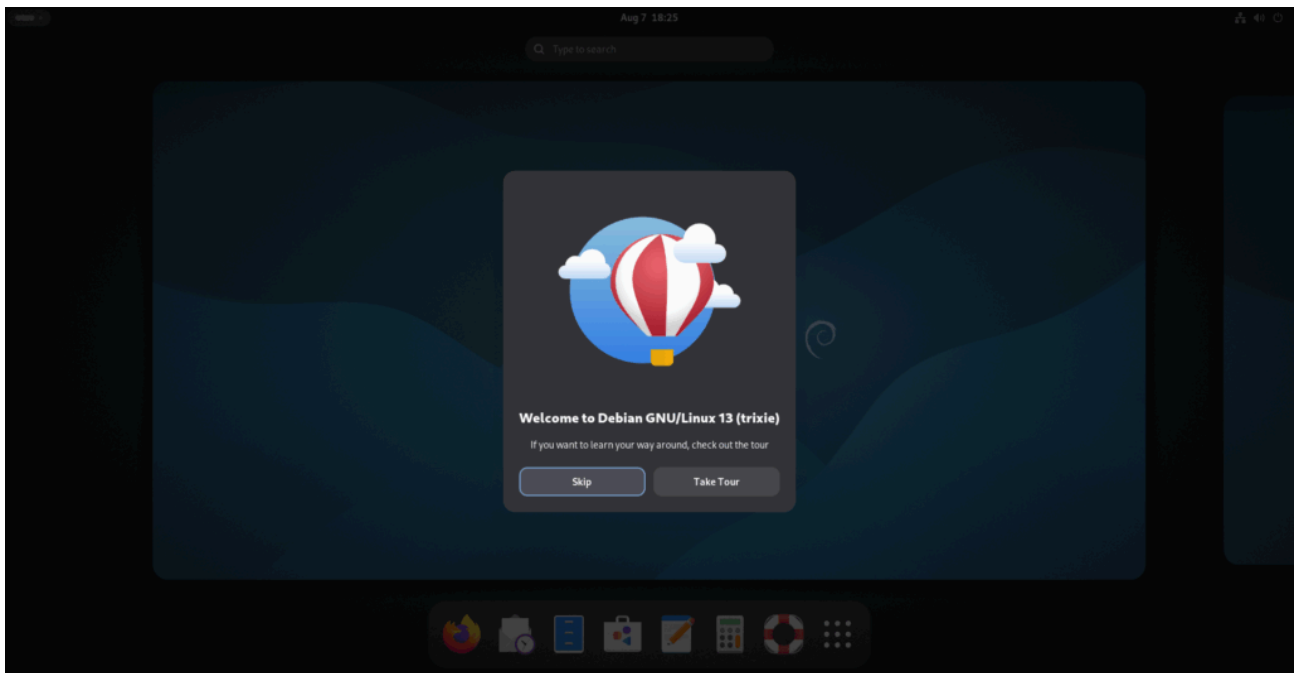


Login to Debian 13 Trixie Desktop

Congratulations! Debian 13, codenamed Trixie, has been successfully installed with the GNOME desktop environment.

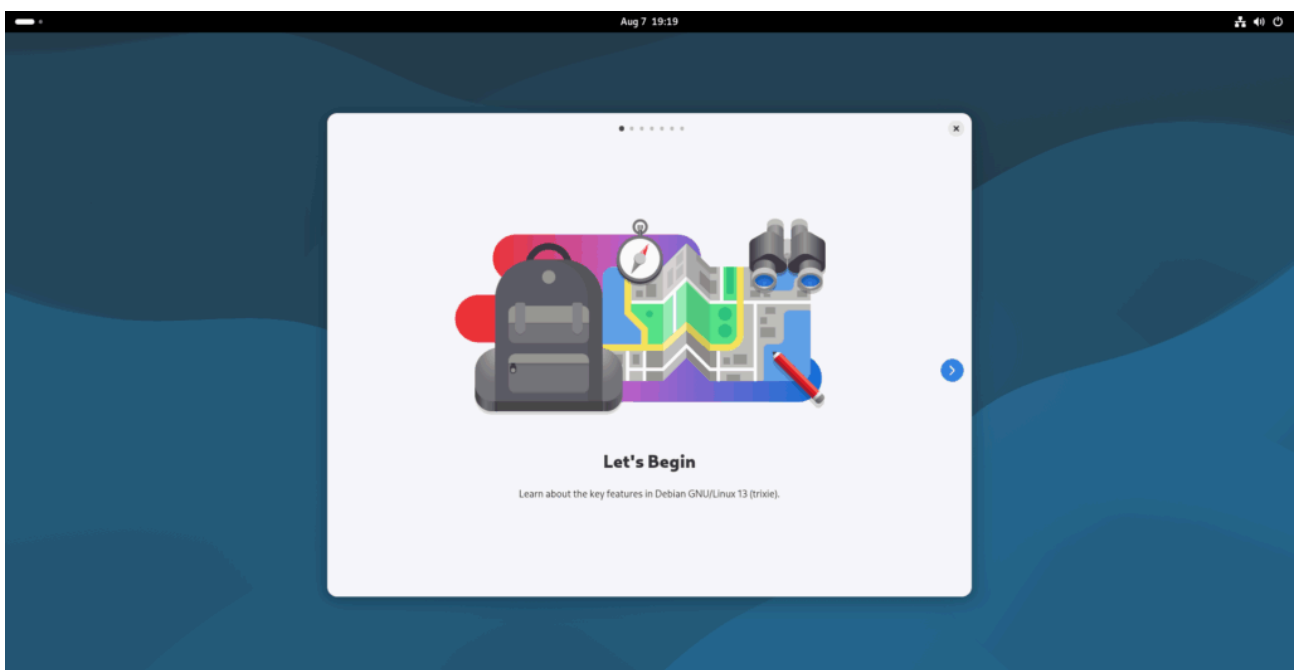
STEP 13: Debian Initial Setup

At first login, you will be greeted with the Gnome tour wizard. If you want to know what's new in the GNOME, click "Take Tour".



GNOME Take Tour Wizard

This will take you on a small tour where you can learn the key features in the new Debian 13 trixie. Press the right/left arrow buttons to go through the wizard.



Key Features in Debian 13

Once done, close the wizard.

That's it. You can now start using the newly installed Debian 13 trixie desktop.



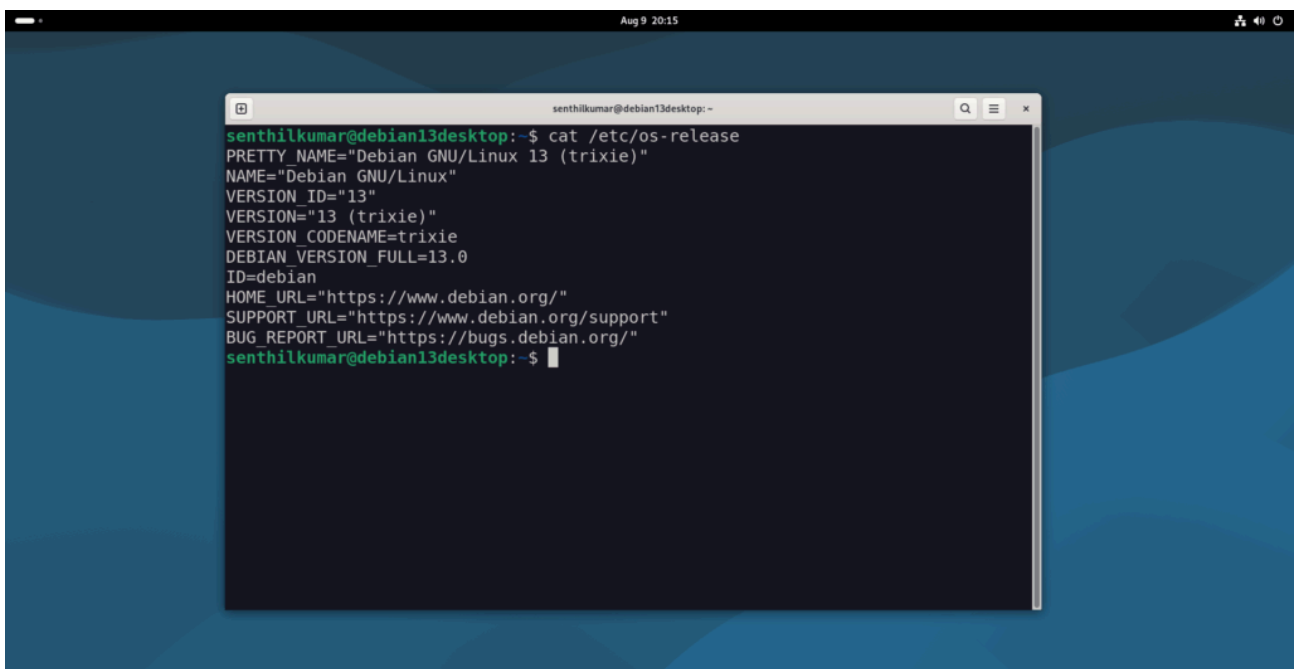
Debian 13 Trixie with GNOME Desktop

STEP 14: Check Debian Version

To verify the installed version of Debian, you can run the following command in the terminal:

```
cat /etc/os-release
```

This command will display information about the Debian release, including the version number, codename, and other relevant details.

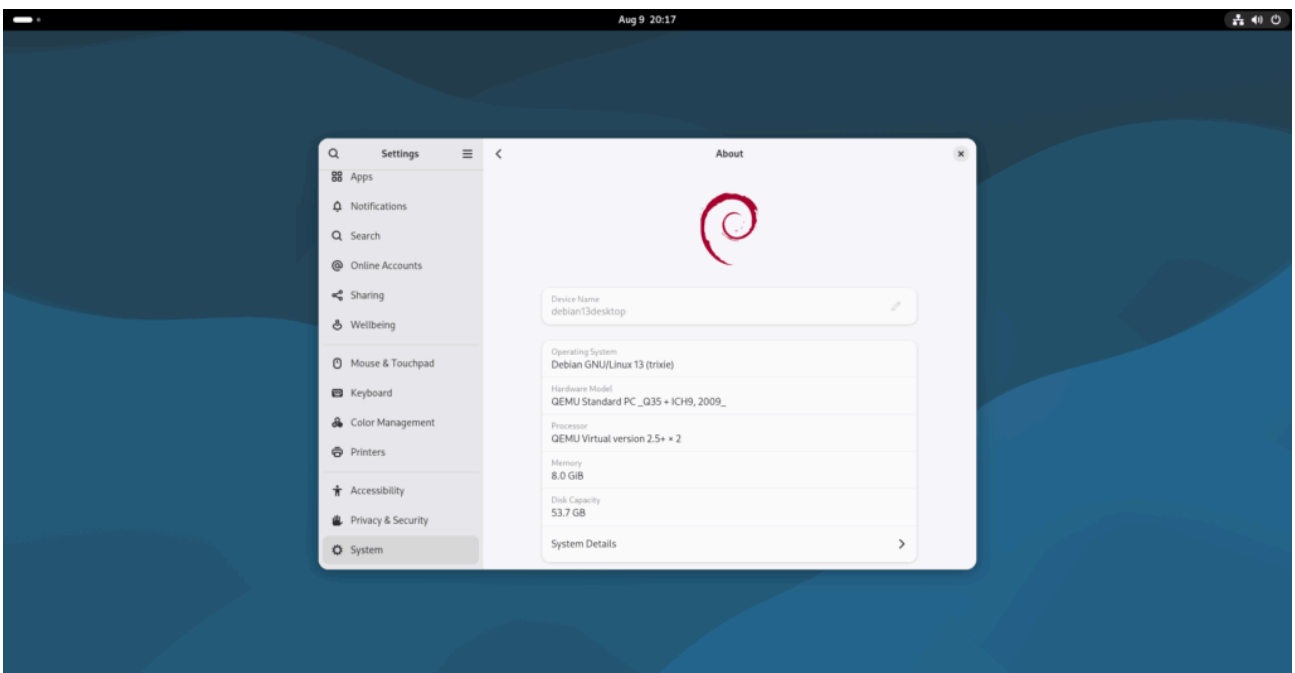


Check Debian 13 Version from Terminal

You can also check the installed Debian version from the Settings section.

To check the Debian version from the GUI settings, you can follow these steps:

1. Click on the "Activities" button located in the top-left corner of the desktop or press the "Super" key (Windows key) on your keyboard.
2. Type "Settings" in the search bar and click on the "Settings" application that appears in the search results.
3. In the Settings window, scroll down and click on the "About" section.
4. The About section will display various system information, including the Debian version. Look for the version number or codename mentioned in this section to identify your Debian release.



Check Debian 13 Version from GUI Settings Window

Conclusion

We have successfully completed the installation of Debian 13, codenamed Trixie. Debian 13 offers a wide range of features and a user-friendly experience, enhanced by the GNOME desktop environment.

With its stability, extensive software repository, and the ability to customize your system, Debian 13 provides a robust platform for both desktop and server environments.

Remember to explore the vast selection of software available through Debian's package manager, [apt](#), to expand the functionality of your system.

Regularly updating your Debian installation ensures that you have the latest security patches and software updates.

Related Read:

- [Debian Minimal GNOME Install: A Step-by-Step Tutorial](#)
- [How To Dual Boot Windows And Debian](#)

How To Upgrade To Debian 13 Trixie From Debian 12 Bookworm

Debian 13 Upgrade Guide: Transitioning from Debian 12 for Improved Performance.

Published: Updated: 9.1K views 15 mins read

Debian 13, codenamed **Trixie**, has been [officially released](#) on August 9, 2025! This Step-by-Step guide explains how to **upgrade to Debian 13 trixie from Debian 12 bookworm** from command line.

From updating repositories to backing up your data, we'll cover all the necessary steps to ensure a **successful Debian 13 upgrade**.

Step-by-Step Guide to Upgrade to Debian 13 from Debian 12

The **release of Debian 13 (trixie)** brings forth a wealth of new features, enhanced performance, and increased stability. **Upgrading your Debian 12 system to Debian 13** will unlock the full potential of this latest operating system.

In this detailed guide, we provide you with an easy-to-follow step-by-step process for seamlessly **upgrading to Debian 13**.

Step 1: Verify Your Current Debian Version

First, you need to confirm that you are currently running Debian 12.

To check the Debian distribution name and version number, run the following command in your terminal:

```
lsb_release -d
```

This should return **"Description: Debian GNU/Linux 12 (bookworm)"** if you're running Debian 12.

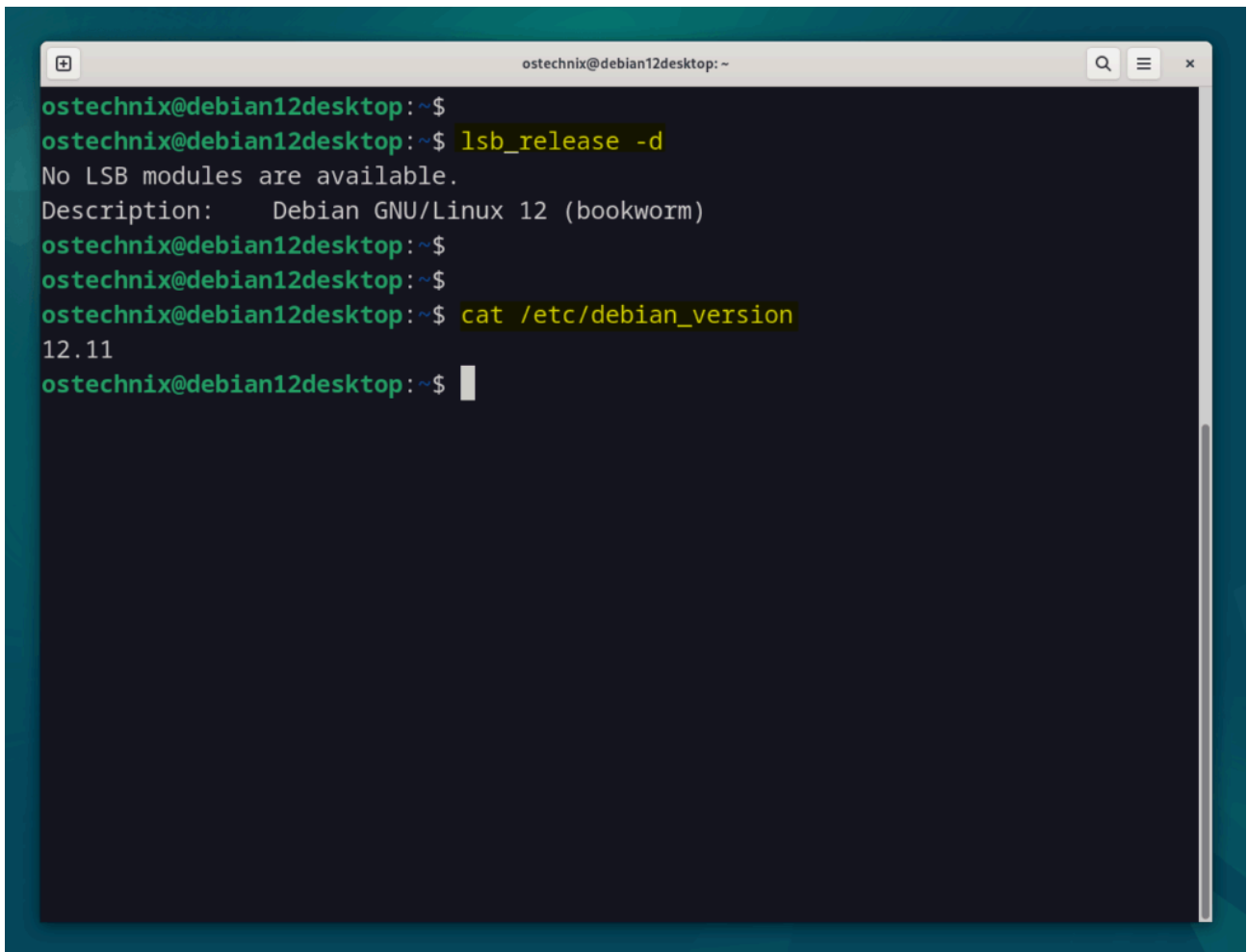
You can also display the version number using the following command:

```
cat /etc/debian_version
```

This should return 12.x where x is the current point release of Debian 12.

Sample output:

12.11

A terminal window titled 'ostechnix@debian12desktop: ~' with search, menu, and close icons in the title bar. The terminal shows the following commands and output:

```
ostechnix@debian12desktop:~$  
ostechnix@debian12desktop:~$ lsb_release -d  
No LSB modules are available.  
Description:    Debian GNU/Linux 12 (bookworm)  
ostechnix@debian12desktop:~$  
ostechnix@debian12desktop:~$  
ostechnix@debian12desktop:~$ cat /etc/debian_version  
12.11  
ostechnix@debian12desktop:~$
```

View Current Debian Version

Step 2: Update and Upgrade Your Debian 12 Packages

Prior to upgrading Debian bookworm to trixie, make sure all existing packages are updated and upgraded to their latest available versions.

Open a terminal window. Run the following commands one by one, entering your sudo (administrator) password when prompted:

```
sudo apt update
```

```
sudo apt upgrade
```

```
sudo apt dist-upgrade
```

Once the upgrades are complete, clean any residual packages and configuration files using the following commands:

```
sudo apt --purge autoremove
```

```
sudo apt autoclean
```

```
sudo find /etc -name '*.dpkg-' -o -name '*.ucf-' -o -name '*.merge-error'
```

With the introduction of numerous new packages, the Debian trixie release discontinues and excludes several old packages that were present in bookworm. These obsolete packages do not have a designated upgrade path.

While it is technically possible to continue using these obsolete packages, it's important to note that the Debian project generally ceases security support for them approximately one year after the release of trixie.

Furthermore, the Debian project typically does not provide other forms of support during this period. Therefore, it is recommended to replace these obsolete packages with available alternatives, if any exist.

To identify and remove "Obsolete and Locally Created Packages" from the command line, you can use the following commands:

To list all obsolete packages, enter:

```
apt list '~o'
```

To remove these obsolete packages, use the purge command with the '~o' filter:

```
sudo apt purge '~o'
```

It is also advisable to purge removed packages along with their configuration files. To list all removed packages that may have configuration files left on the system, run:

```
apt list '~c'
```

Remove those leftovers using command:

```
sudo apt purge '~c'
```

These commands will help ensure that your system remains clean and free from unnecessary packages after the upgrade process.

To apply all the updates, it is recommended to reboot your system. To do so, execute the following command:

```
sudo reboot
```

After the system restarts, you can proceed with the Debian bookworm to trixie upgrade process.

Step 3: Backup your Data

The next most important step is to create a backup of your data.

Make sure you have a copy of all important personal files, directories, and configuration files before proceeding with the upgrade process.

Your backup serves as a safety net to ensure the preservation of your important data in case of any unforeseen issues during the upgrade process.

To perform a comprehensive backup, focus on backing up the following:

1. **The contents of the /etc directory:** This directory contains various system configuration files. Make sure to create a backup of its entire contents.
2. **The /var/lib/dpkg directory:** This directory holds the package database. It is important to have a backup of this directory to preserve package information.
3. **The /var/lib/apt/extended_states file:** This file contains additional information about installed packages. Include this file in your backup to retain relevant package states.
4. **The output of the dpkg --get-selections "*" command:** Execute this command, including the quotes, to generate a list of all installed packages. Backing up this output will help you restore your package selections.

```
# Set timestamp variable for consistency
TS=$(date +%Y%m%d-%H%M%S)

# Create a backup directory
BACKUP_DIR="debian-backup-$TS"
mkdir -p "$BACKUP_DIR"

# 1. Backup the entire /etc directory
sudo tar -czvf "$BACKUP_DIR/etc-backup-$TS.tar.gz" /etc

# 2. Backup the /var/lib/dpkg directory
sudo tar -czvf "$BACKUP_DIR/dpkg-backup-$TS.tar.gz" /var/lib/dpkg

# 3. Backup the /var/lib/apt/extended_states file
sudo cp /var/lib/apt/extended_states "$BACKUP_DIR/extended_states-$TS.backup"

# 4. Save the list of all installed packages
dpkg --get-selections "*" > "$BACKUP_DIR/package-selections-$TS.list"
```

The resulting folder structure would look like below:

```
debian-backup-20260316-1742/
├─ etc-backup-20260316-1742.tar.gz
├─ dpkg-backup-20260316-1742.tar.gz
├─ extended_states-20260316-1742.backup
└─ package-selections-20260316-1742.list
```

Here's what it does in order:

1. Creates a debian-backup directory.

2. Compresses and saves `/etc` as `etc-backup.tar.gz`.
3. Compresses and saves `/var/lib/dpkg` as `dpkg-backup.tar.gz`.
4. Copies `extended_states` into the backup folder.
5. Saves the list of installed packages into `package-selections.list`.

As you see, each backup has the date and time in its filename. That way, you can keep multiple backups without overwriting.

By backing up these essential components, you can safeguard your data and configuration settings. This ensures that you have the necessary resources to restore your system or recover specific files if needed.

You also have a variety of options when it comes to selecting [backup applications](#) for your Debian Linux system. Consider using one or two of the following applications to backup your data and system settings:

1. [Rsync](#): This powerful command-line utility is ideal for backing up personal data and the contents of your `$HOME` directory.
2. [Deja Dup](#): If you prefer a simple graphical interface, Deja Dup is a user-friendly utility that allows you to back up your personal data with ease.
3. [Timeshift](#): For backing up system files and settings, Timeshift provides a graphical application that enables you to create snapshots of your system.
4. [Restic](#): This command-line utility offers the ability to save multiple revisions of files and directories in an encrypted repository stored on different backends.
5. [Rsnapshot](#): If you require periodic snapshots of both local and remote machines over SSH, Rsnapshot is a command-line utility that can fulfill this requirement.

Personally, I use **Deja Dup** and **Timeshift** for my backups. Feel free to choose any combination of the above-listed applications that best suits your preferences and backup needs.

Step 4: Modify the `/etc/apt/sources.list` File

Prior to updating the software repositories in Debian bookworm, please back up the current software source list. Follow these steps to create a backup of the current software repository configuration files.

Create a new directory named **"apt"** in your home directory by running the following command:

```
mkdir ~/apt
```

Make a copy of the main software source list by executing the following command with administrator privileges:

```
sudo cp /etc/apt/sources.list ~/apt
```

Create a copy of the additional software sources located in the "**sources.list.d**" directory using the recursive and verbose options:

```
sudo cp -rv /etc/apt/sources.list.d/ ~/apt
```

To ensure that the software repository lists have been successfully copied, verify the contents of the "~/apt" directory by running the following command:

```
$ ls apt/  
sources.list  sources.list.d
```

After executing these steps, you should see the "sources.list" file and the "sources.list.d" directory listed in the output, confirming that the backup process was completed successfully.

Next, you'll need to update your package sources to point to the Debian 13 trixie repositories. To update the Debian 12 (bookworm) repositories to Debian 13 (trixie) repositories, use the following commands.

From the terminal, run the following command with administrator privileges to update the main software source list:

```
sudo sed -i 's/bookworm/trixie/g' /etc/apt/sources.list
```

This command uses the sed utility to replace all occurrences of "bookworm" with "trixie" in the /etc/apt/sources.list file.

Update the additional software source lists in the "sources.list.d" directory by executing the following command:

```
sudo sed -i 's/bookworm/trixie/g' /etc/apt/sources.list.d/*
```

Similarly, this command replaces "bookworm" with "trixie" in all files within the /etc/apt/sources.list.d/ directory.

By following these steps, you will have replaced and updated the Debian 12 (bookworm) repositories with the Debian 13 (trixie) repositories in your system's software source lists.

Verify if the sources list file has been updated with the new repository links by running the following command:

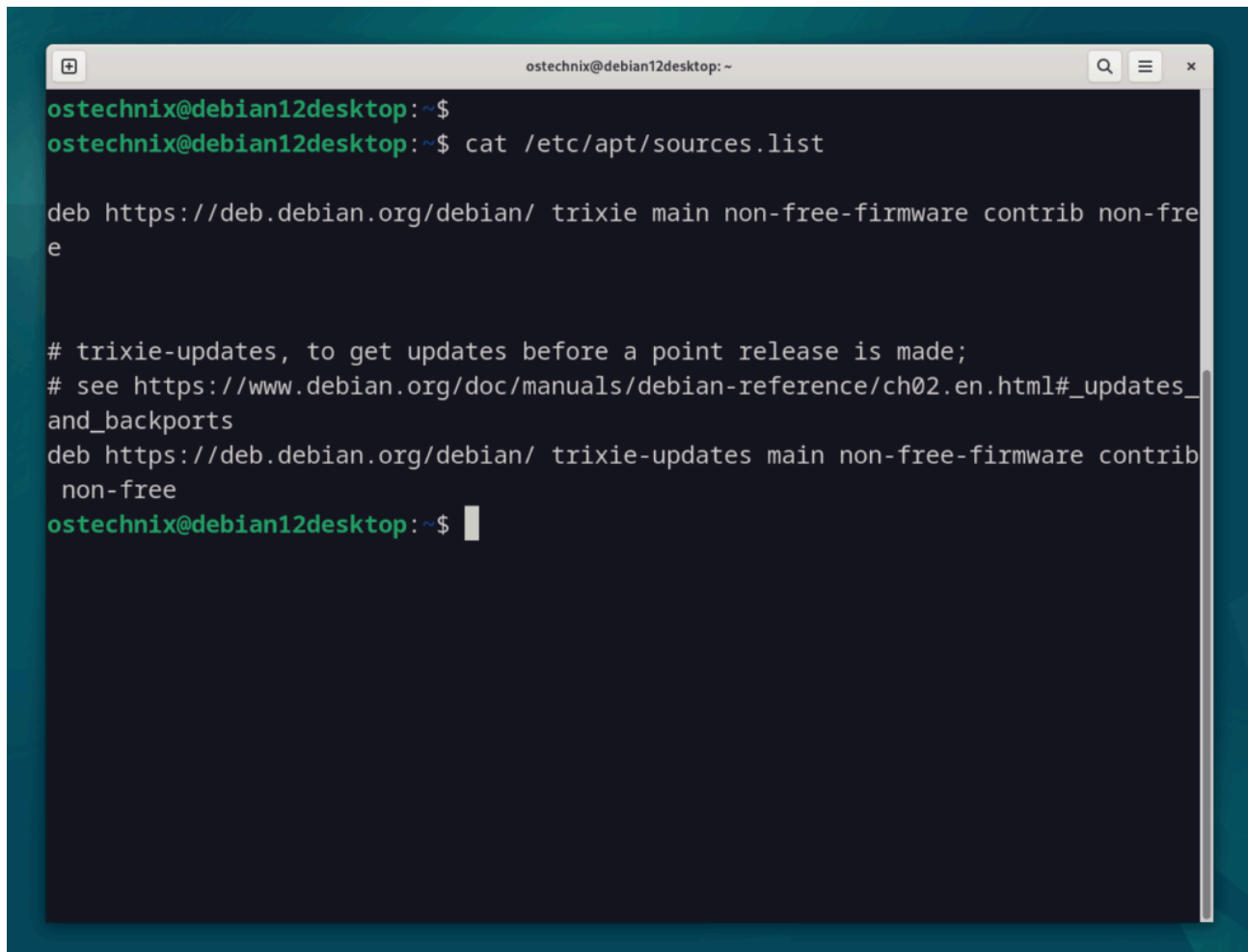
```
cat /etc/apt/sources.list
```

This command displays the contents of the /etc/apt/sources.list file, allowing you to confirm if the repository links have been successfully updated.

Here is the contents of my test machine's source repository list:

```
deb https://deb.debian.org/debian/ trixie main non-free-firmware contrib non-free
```

```
# trixie-updates, to get updates before a point release is made;  
# see https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#_updates_and_backports  
deb https://deb.debian.org/debian/ trixie-updates main non-free-firmware contrib  
non-free
```

A terminal window titled 'ostechnix@debian12desktop: ~' with search, menu, and close icons in the title bar. The terminal shows the command 'cat /etc/apt/sources.list' being executed. The output displays the contents of the file, including the trixie-updates repository entry and a comment about getting updates before a point release.

```
ostechnix@debian12desktop: ~$  
ostechnix@debian12desktop: ~$ cat /etc/apt/sources.list  
  
deb https://deb.debian.org/debian/ trixie main non-free-firmware contrib non-free  
e  
  
# trixie-updates, to get updates before a point release is made;  
# see https://www.debian.org/doc/manuals/debian-reference/ch02.en.html#_updates_  
and_backports  
deb https://deb.debian.org/debian/ trixie-updates main non-free-firmware contrib  
non-free  
ostechnix@debian12desktop: ~$
```

Debian 13 Sources List

As observed in the above output, the **sources.list** file has been successfully updated with the new Debian 13 repository links.

Next, update the repository lists by executing the following command in the terminal:

```
sudo apt update
```

This command will refresh the repository lists based on the updated configuration in the **sources.list** file. It allows your system to retrieve the latest package information from the newly added Debian 13 repositories.

Step 5: Perform Minimal System Upgrade

In certain scenarios, performing a full system upgrade as explained in the next step may result in the removal of a significant number of packages that you wish to retain.

Therefore, Debian developers recommend a two-part upgrade process consisting of a minimal upgrade followed by a full upgrade.

The minimal upgrade ensures the upgrade of all existing packages without installing or removing any additional packages.

To initiate the minimal upgrade, use the following command:

```
sudo apt upgrade --without-new-pkgs
```

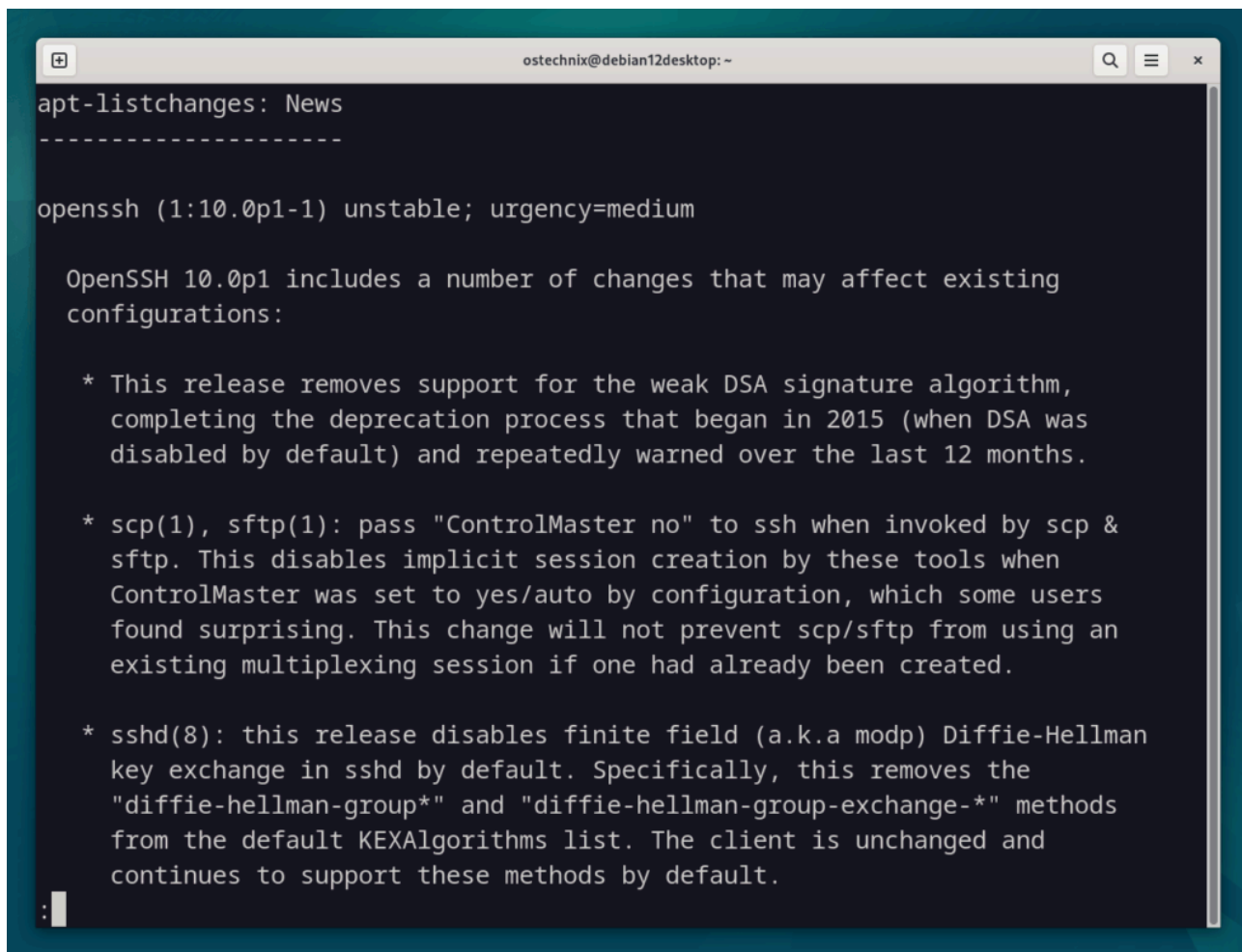
While the upgrade process is underway, carefully monitor the screen. You may encounter a series of questions that require your input, such as deciding the fate of a configuration file or whether to restart a particular service.

The specific questions presented to you can vary based on the number of installed packages on your current Debian system.

You should pay attention and respond to these prompts appropriately to ensure a smooth upgrade experience. This allows you to retain desired packages and maintain the desired configuration during the minimal upgrade phase.

For example, if you have the **apt-listchanges** package installed, it will display important information about upgraded packages in a pager after downloading them.

To access the information, press ENTER to read through the details. If you wish to exit and return to the upgrade process, simply press the **q** key.

A terminal window titled 'ostechnix@debian12desktop: ~' with search, menu, and close icons in the title bar. The terminal output shows the command 'apt-listchanges: News' followed by a dashed line separator. Below this, it shows 'openssh (1:10.0p1-1) unstable; urgency=medium'. A paragraph follows: 'OpenSSH 10.0p1 includes a number of changes that may affect existing configurations:'. This is followed by three bullet points: 1. 'This release removes support for the weak DSA signature algorithm, completing the deprecation process that began in 2015 (when DSA was disabled by default) and repeatedly warned over the last 12 months.' 2. 'scp(1), sftp(1): pass "ControlMaster no" to ssh when invoked by scp & sftp. This disables implicit session creation by these tools when ControlMaster was set to yes/auto by configuration, which some users found surprising. This change will not prevent scp/sftp from using an existing multiplexing session if one had already been created.' 3. 'sshd(8): this release disables finite field (a.k.a modp) Diffie-Hellman key exchange in sshd by default. Specifically, this removes the "diffie-hellman-group*" and "diffie-hellman-group-exchange-*" methods from the default KEXAlgorithms list. The client is unchanged and continues to support these methods by default.' A cursor is visible at the bottom left of the terminal window.

```
ostechnix@debian12desktop: ~
apt-listchanges: News
-----

openssh (1:10.0p1-1) unstable; urgency=medium

OpenSSH 10.0p1 includes a number of changes that may affect existing
configurations:

* This release removes support for the weak DSA signature algorithm,
  completing the deprecation process that began in 2015 (when DSA was
  disabled by default) and repeatedly warned over the last 12 months.

* scp(1), sftp(1): pass "ControlMaster no" to ssh when invoked by scp &
  sftp. This disables implicit session creation by these tools when
  ControlMaster was set to yes/auto by configuration, which some users
  found surprising. This change will not prevent scp/sftp from using an
  existing multiplexing session if one had already been created.

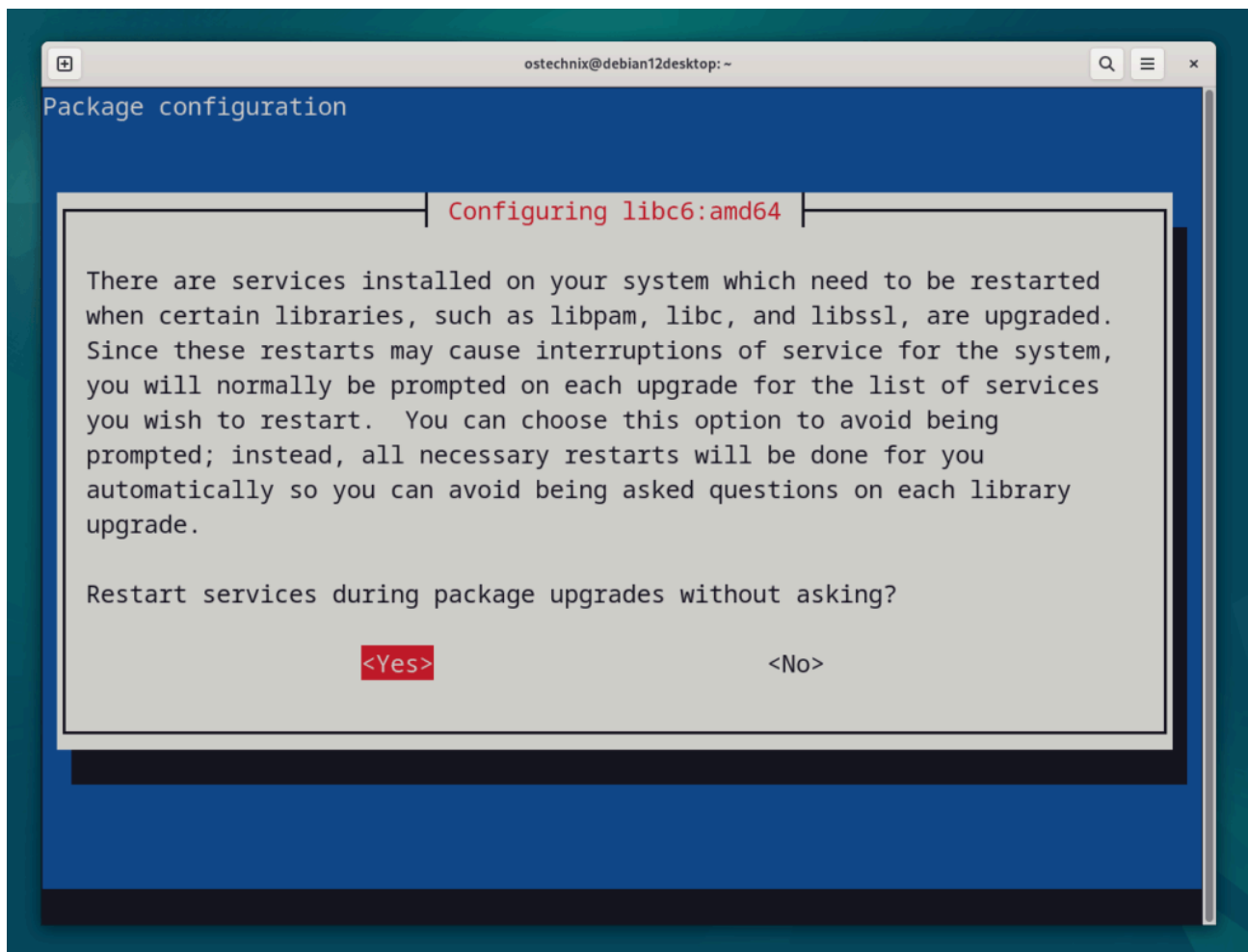
* sshd(8): this release disables finite field (a.k.a modp) Diffie-Hellman
  key exchange in sshd by default. Specifically, this removes the
  "diffie-hellman-group*" and "diffie-hellman-group-exchange-*" methods
  from the default KEXAlgorithms list. The client is unchanged and
  continues to support these methods by default.
```

Apt-listchanges News

During the upgrade process, various services installed on your Debian system may require restarting when specific libraries like `libpam`, `libc`, and `libssl` are upgraded. Restarting these services may result in temporary interruptions of service.

To minimize these interruptions and avoid being prompted for each library upgrade, you have the option to choose a default action.

If you prefer not to be asked about restarting services individually, you can select **"Yes"** and press ENTER to proceed. This choice ensures that the upgrade process continues without interruption and automatically restarts the necessary services as needed.



Choose Yes To Restart Services During Package Upgrades Without Asking

The minimal upgrade process will be typically completed within a few minutes.

Step 6: Upgrade to Debian 13

Now that your package sources have been updated, you can upgrade your system.

Execute the following command to perform a full system upgrade to Debian 13 trixie:

```
sudo apt full-upgrade
```

Once again, it is important to pay attention to the screen during the upgrade process and respond to any prompts or questions that may arise.

As in the minimal upgrade phase, you will see essential information about the upgraded packages (if `apt-listchanges` is installed) in a pager. Press **q** to exit and continue the upgrade.

Some services may need to restart when libraries like `libpam`, `libc`, or `libssl` are upgraded, causing brief interruptions.

To avoid prompts for each restart, select **"Yes"** and press **ENTER**. This lets the upgrade proceed without interruption and restarts services automatically.

After completing the upgrade process to Debian 13 (trixie), it is recommended to reboot your system to ensure that all changes take effect.

To reboot your system, run:

```
sudo reboot
```

Once your system restarts and you log in, you can verify the Debian 13 trixie version using either of the following commands:

To check the version by reading the contents of the `/etc/debian_version` file, run:

```
$ cat /etc/debian_version
13.4
```

This command will display the version information, which should indicate **"13.x"** if the upgrade to Debian 13 (trixie) was successful.

If you want detailed output, use this command instead:

```
cat /etc/os-release
```

Sample Output:

```
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"
NAME="Debian GNU/Linux"
VERSION_ID="13"
VERSION="13 (trixie)"
VERSION_CODENAME=trixie
DEBIAN_VERSION_FULL=13.4
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
```

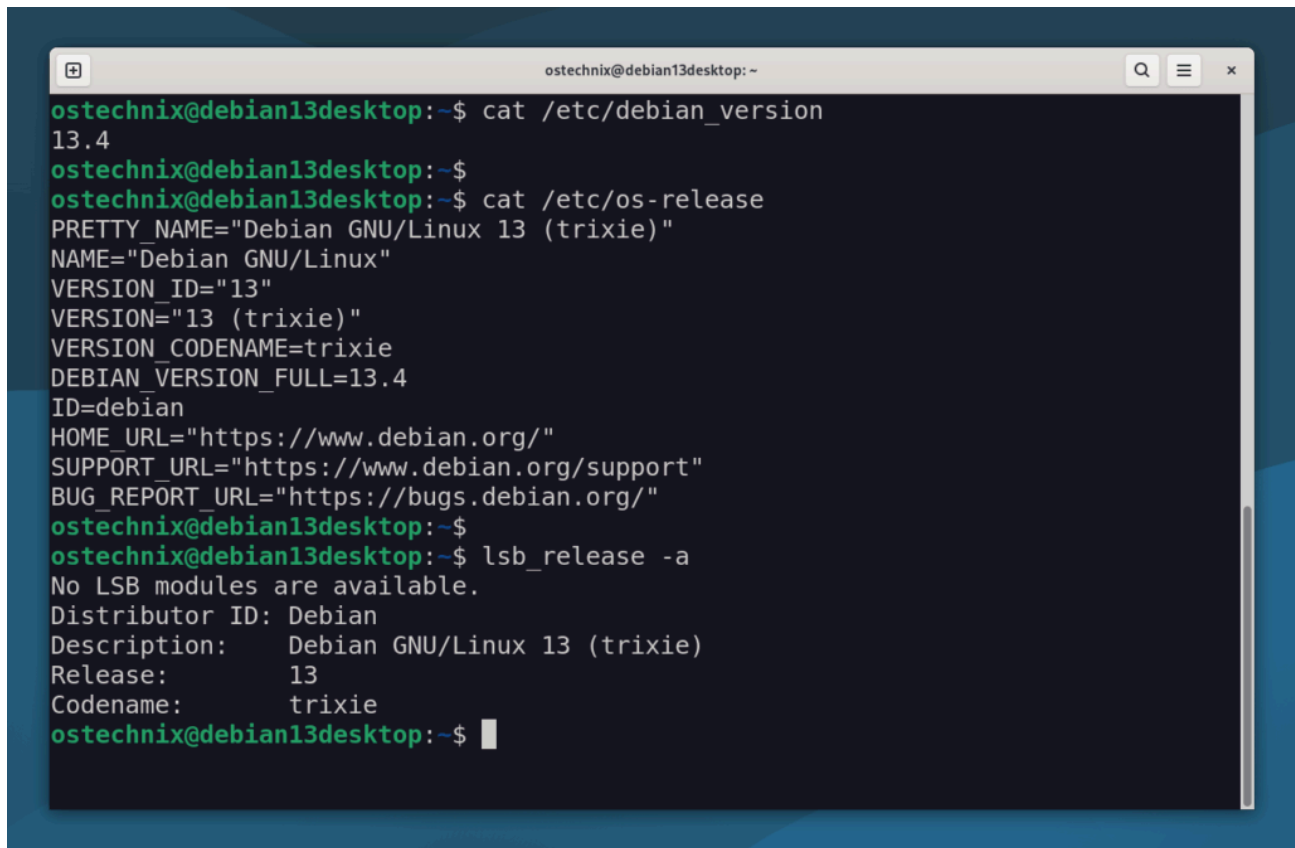
Alternatively, you can use the `lsb_release` command to obtain detailed information about the Debian release:

```
lsb_release -a
```

This command will provide comprehensive information, including the distribution ID, release number, codename, and more. Review the output to confirm that the Debian version is correctly identified as "trixie".

Sample Output:

```
No LSB modules are available.
Distributor ID:    Debian
Description:      Debian GNU/Linux 13 (trixie)
Release:         13
Codename:        trixie
```

A terminal window titled 'ostechnix@debian13desktop: ~' with search and menu icons in the top right. The terminal shows the following commands and output:

```
ostechnix@debian13desktop:~$ cat /etc/debian_version
13.4
ostechnix@debian13desktop:~$ 
ostechnix@debian13desktop:~$ cat /etc/os-release
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"
NAME="Debian GNU/Linux"
VERSION_ID="13"
VERSION="13 (trixie)"
VERSION_CODENAME=trixie
DEBIAN_VERSION_FULL=13.4
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
ostechnix@debian13desktop:~$ 
ostechnix@debian13desktop:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Debian
Description:    Debian GNU/Linux 13 (trixie)
Release:       13
Codename:      trixie
ostechnix@debian13desktop:~$
```

Check Debian 13 Version From Commandline

Congratulations on successfully upgrading Debian 12 (bookworm) to Debian 13 (trixie)!

To ensure a clean system and remove any packages that are no longer required, you can execute the following commands.

Remove all packages that are no longer needed, including their configuration files:

```
sudo apt --purge autoremove
```

This command will remove any unused packages from your system, freeing up disk space and eliminating unnecessary dependencies.

Clean up residual package files that are no longer required:

```
sudo apt autoclean
```

This command will remove cached package files from the system's package cache directory, helping to free up additional disk space.

Step 7: Verify if Everything Works Fine

After logging in to your newly upgraded Debian 13 system, check if everything is functioning correctly. Take the time to open and/or run various installed applications and verify that they are working as expected.

Ensure that all essential functionalities and features are operational. If you encounter any issues or if something breaks your system, please share your issues in the comment section below or seek assistance from the Debian forum.

Keep in mind that Debian 13 has just been released (August 9, 2025), so minor glitches might occur.

However, in my personal experience, the **Debian 13 upgrade process was smooth**, and all aspects of the system worked seamlessly.

Step 8: Restore the Backup (Optional)

If the upgrade process to Debian 13 (trixie) is smooth and error-free, it should not delete any important files.

However, as a precautionary measure, if you find any missing data or files, restore them from the backup you created before the upgrade.

Follow the appropriate procedure to restore your backup files, directories, and configuration settings. This step ensures that your system is set up exactly as it was before the upgrade, including any personalized configurations or data.

Additionally, you can take this opportunity to install new applications of your choice and start exploring the features and enhancements offered by the Debian 13 trixie edition.

That's it. Enjoy using Debian 13 and make the most out of its capabilities!

TL;DR

Here are the key steps to upgrade to Debian 13 (trixie):

1. Update and upgrade all currently installed packages using `sudo apt update`, `sudo apt upgrade`, `sudo apt dist-upgrade`, `sudo apt autoremove`, `sudo apt autoclean`, and `sudo find /etc -name '*.dpkg-' -o -name '*.ucf-' -o -name '*.merge-error'`.
2. Backup all important personal files, directories, and configuration files, such as `/etc`, `/var/lib/dpkg`, `/var/lib/apt/extended_states`, and the output of `dpkg --get-selections "*" command`.
3. Copy the current software source list and `sources.list.d` directory using `sudo cp /etc/apt/sources.list ~/apt` and `sudo cp -rv /etc/apt/sources.list.d/ ~/apt`.
4. Update the `sources.list` file and `sources.list.d` files to replace "bookworm" with "trixie" using `sudo sed -i 's/bookworm/trixie/g' /etc/apt/sources.list` and `sudo sed -i 's/bookworm/trixie/g' /etc/apt/sources.list.d/*`.

5. After modifying the software sources list, run `sudo apt update` command to refresh the software repository list.
6. Perform a minimal upgrade using `sudo apt upgrade --without-new-pkgs` and respond to any prompts or questions that may arise during the upgrade process.
7. Start the main upgrade using `sudo apt full-upgrade` to upgrade all packages and dependencies.
8. Reboot your system after the upgrade process is completed with `sudo reboot`.
9. Verify the Debian 13 trixie version using `cat /etc/debian_version` or `lsb_release -a`.
10. Optionally, remove packages that are no longer required using `sudo apt --purge autoremove` and `sudo apt autoclean`.
11. Log in to your upgraded system, check if everything works fine, and report any issues.
12. If there are any data missing, restore the backup you took before upgrading the system.
13. Install new applications of your choice and start using Debian 13 trixie edition!

Congratulations! You've successfully upgraded the Debian 12 bookworm to Debian 13 trixie.

Resource:

[Debian Official Documentation](#)